

The Cost Geometry of Belief

finite-resource inference under noisy observation

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Abstract. A finite agent — a machine’s digital twin, or any bounded reasoner — sees a fixed, noisy world only through finite sensors, so its coherent output is not a point but a *belief*: a probability density over states (the Bayes posterior). Certainty is denied *twice* — by observation (Cramér–Rao) and by physics (Landauer) — both diverging at the boundary where the Fisher information blows up. We turn this finiteness into geometry: *belief-cost geometry*, the geometry of what it costs to change one’s mind. The cost metric is optimal transport in Wasserstein space, conformally reweighted by Fisher information (the price of the precision at stake), $\tilde{g}_{e,U} = 2(e + U) g_{\mathcal{W}_2}$ with relief U . It rests on *two* posed postulates: that revision *cost* is a *scalar price on transport* (the arena), and that the price is *honest* — one nat costs the same length everywhere (eikonal). Honesty selects the Fisher reweighting because transport *demotes* the Fisher information from the metric ruler of distinguishability (its role in Fisher–Rao information geometry) to the *slope* of entropy. Three results follow on the conformal class: a *wall* — a well-posed inference pushes certainty to infinite cost-distance once the relief dominates the Fisher information (necessity conjectured beyond power laws); an *honest family* — the eikonal price is *equivalent* to proportionality $U = cJ$, the Fisher family; and a *rigidity* (essentially location-scale) — these geometries are hyperbolic and the Stam bound crowns the Gaussian as the *most hyperbolic* location-scale belief (ranking taken at $e = 0$, where K is constant), the value $-1/4$ being one image of a *relativity of cost*. The cost of reaching a given *precision* then has a geometric floor diverging at certainty. Thermodynamics fixes the cost unit (one nat costs $k_B T$ at the wall) and motivates the framework; the results are geometric, in nats.

Keywords. belief-cost geometry; optimal transport; Wasserstein space; Fisher information; information geometry; belief revision; eikonal equation; Stam inequality; Gaussian; thermodynamics of inference; entropy.

1 Introduction

A *digital twin* (Grievens and Vickers, 2017) maintains, inside a machine, an image of a real system — a territory, a city, a structure — in order to track or anticipate it. It is often conceived as a *replica*: a deterministic copy, regularly refreshed with data. In the setting we consider, however, the machine that realizes it is *finite* and accesses the system only through finite, noisy channels — whether a physical sensor or a *finite agent* (a human included), each itself noisy. Its coherent output is then a *belief*: a probability density over the possible states, produced by Bayesian inference, the coherent framework for reasoning under uncertainty (Cox, 1946; Jaynes, 2003; MacKay, 2003). Practice already does this — ensembles, data assimilation, Kalman filters (Kalman, 1960), often implicitly. Figure 1 shows this acquisition pipeline — a fixed hidden state, a finite noisy sensor, and the Bayes posterior it yields.

To such a belief, *certainty* — the perfect twin, without uncertainty — is denied *twice*. First by observational finiteness: perfect precision would require unbounded information, which the

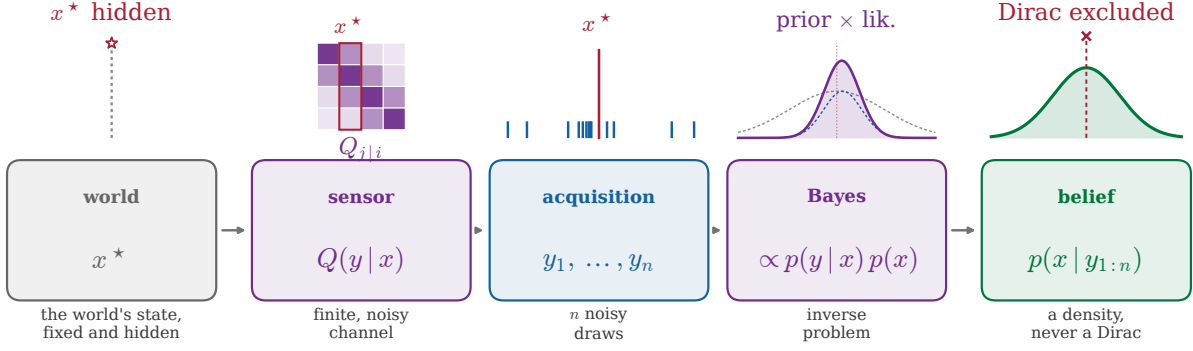


Figure 1: The inference chain: from a noisy world to a belief. The state x^* (fixed, hidden); the finite and noisy sensor $Q(y|x)$ ($Q_{j|i} = P(y=b_j | x=a_i)$); the n draws $y_1, \dots, y_n$, scattered around x^* by the noise; the belief $p(x|y_{1:n}) \propto p(y|x)p(x)$ obtained by Bayes. The output is a *density* (of entropy $H = -\int p \ln p$); certainty, the barred centered Dirac, is excluded from it.

Cramér–Rao inequality rules out (Cover and Thomas, 2006, Th. 11.10.1). Then by physical finiteness: erasing information dissipates energy, by the Landauer principle (Landauer, 1961). Both refusals diverge at the boundary where the Fisher information J blows up (Fisher, 1925). The belief is born of inference; finiteness only keeps it from freezing into certainty — a boundary no finite system crosses, which is also a *reprieve*. Lying at infinite cost-distance, the wall makes certainty not only unreachable but, by the symmetry of that distance, just as far in return: a Dirac sits infinitely far from every interior belief, so certainty — could it be held — would be beyond revision, dogmatic, beyond Bayesian repair by Cromwell’s rule (Lindley, 1991). A finite agent therefore lives in the interior, where every belief is a finite revision from any other — never frozen, never over-committed; keeping some uncertainty is what keeps it adaptable.

The belief thus laid down, what remains to describe is *geometric*: the shape of the space where these beliefs live and move — its distances, its *boundary* (certainty, the excluded Dirac) and its curvature. A geometric toolbox already exists, and each piece, in its own way, meets that boundary: information geometry equips beliefs with the Fisher–Rao metric (Chentsov, 1972; Amari, 2016), under which the Dirac lies at *infinite* distance — but what it measures is *distinguishability*, not cost; optimal transport metrizes the displacement of mass (Otto, 2001), yet there the Dirac is at *finite* distance, with no wall; the thermodynamics of information prices the erasure of certainty (Landauer) and the dissipation of a driven belief (Sivak and Crooks, 2012; Aurell et al., 2011; Ito, 2023), but on a *given* geometry; and Bayesian mechanics ties a dissipative agent to a geometry of belief (Friston, 2019; Sakthivadivel, 2022), without a wall. To our knowledge, none re-metrizes beliefs into a geometry of revision *cost* in which certainty is an *uncrossable boundary* — a wall at infinite cost-distance — carrying a curvature; that is what we characterize. Where Fisher–Rao is the canonical geometry of *distinguishability* (Čencov), this is, *we argue*, its dual for *revision cost* — canonical within its class (theorem-grade on the power family; the general case is Conjecture C1, remark 6.1), as Čencov’s is within his. This thesis organizes the paper (remark 7.1).

Plan. Section 2 sketches the proposed formulation in broad strokes; Section 3 makes it formal (the arena, the two postulates, the cost class); Section 4 proves the characterizations — the wall, the eikonal $U = cJ$, and hyperbolicity with Stam rigidity; Section 5 draws from them a well-posed inference with a cost floor; Section 6 delimits scope and open problems and Section 7 situates the work; the machinery proofs are deferred to Appendix A.

2 Proposed formulation

Problem 1. Characterize the geometry of the space of beliefs that a finite agent, inferring a fixed and noisy world through finite sensors, can inhabit: its distances (the cost of changing one's mind), its boundary (certainty) and its curvature.

We pose two postulates (made precise in Section 3): **P0** (the arena) — a revision cost is a scalar price on optimal transport, so beliefs live in the Wasserstein space $(\mathcal{P}_2, \mathcal{W}_2)$ — $\mathcal{P}_2$ the densities of finite spread (the possible posteriors), $\mathcal{W}_2$ the optimal-transport distance between two beliefs (the least cost of morphing one into the other) — whose infinitesimal ruler is the transport metric $g_{\mathcal{W}_2}$ (Otto, 2001; Ambrosio et al., 2008); and **P1** (honesty) — the price is *eikonal*, one nat of knowledge costing the same length everywhere. Their consequence is a single cost metric, the transport metric conformally reweighted by the Fisher information:

$$\tilde{g}_{e,U} = 2(e + U) g_{\mathcal{W}_2}, \quad U = cJ, \quad J = \|\nabla_{\mathcal{W}_2} H\|^2 = \int \frac{|\nabla p|^2}{p}$$

where p is the belief (a density over states $x \in \mathbb{R}^n$, n the state dimension), $H = -\int p \ln p$ its entropy (in nats) and $-H$ its knowledge; $\tilde{g}_{e,U}$ is the resulting cost metric — the local ruler giving the price of a small revision, namely the transport ruler $g_{\mathcal{W}_2}$ reweighted point by point by $2(e + U)$ — with relief $U \geq 0$ the local price of precision, $e \geq 0$ a baseline, and $c > 0$ a fixed cost unit (only cost ratios mean anything; $c = 2$ below is one representative). The two gradients differ: ∇p is the ordinary spatial gradient, so $J = \int |\nabla p|^2 / p$ is the classical Fisher information (large where the belief is sharp, infinite at certainty), whereas $\nabla_{\mathcal{W}_2} H$ is the gradient of H taken along transport (Otto's calculus on $\mathcal{P}_2$), its squared length equal to J by the Otto identity (Otto, 2001; Villani, 2009). That identity is the hinge: along transport the Fisher information is the squared slope of $-H$, not the distinguishability ruler it is in Fisher–Rao — the demotion that lets honesty (P1) select $U = cJ$ and threads the paper, every result re-reading as a fact about $-H$.

From the two postulates the characterization follows, on the conformal class (certainty being the limit where the belief sharpens to a point):

- a **wall** (theorem 4.10): a well-posed inference (existence and stability; uniqueness left open) rejects certainty to infinite cost-distance as soon as the relief dominates the Fisher information (sufficiency in every dimension; necessity theorem-grade on the power family $\{cJ^\alpha\}$ (power-law reliefs), the general case conjectured, remark 6.1);
- an **honest family** (theorem 4.15): the eikonal price is equivalent to $U = cJ$, the Fisher family;
- a **rigidity** (theorems 4.18 and 4.19), essentially location-scale (a fixed base shape positioned by a centre μ and scaled by a spread σ): these geometries are hyperbolic (negative Gauss curvature $K = -1/(2cJ_0)$), where J_0 is the Fisher information of the family's standardized shape (the variance-1 base; distinct from the precision $J = J_0/\sigma^2$); the Stam bound $J_0 \geq 1$ (equality iff Gaussian) crowns the Gaussian as the most hyperbolic location-scale belief (ranking at $e = 0$) ($-\frac{1}{4}$ being its image under a unit).

These split along the two roles of the entropy: its value $-H$ (how concentrated the belief is) sets the wall and the cost floor, while the base slope $\sqrt{J_0}$ (the family's shape) sets the curvature $K = -1/(2cJ_0)$ — so the Gaussian's extremality (a J_0 fact) is orthogonal to the wall (a $-H$ fact). Whence a cost floor for reaching a given precision, diverging at certainty, and a relativity of cost: only relative cost means anything, the value $-\frac{1}{4}$ being one image under a change of unit (theorem 4.23). Thermodynamics fixes that unit (one nat costs $k_B T$, the thermal energy scale, at the wall) and motivates the picture; the results are geometric, in nats. In one breath: one quantity, the entropy; two postulates — arena and honesty; three faces — the wall, Fisher, Stam.

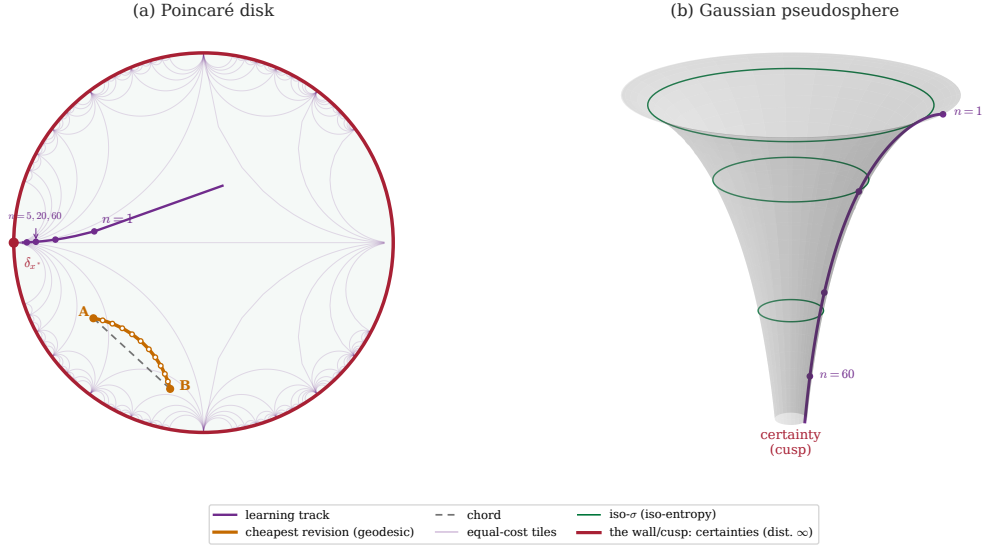


Figure 2: The cost geometry of belief. (a) The location-scale leaf in the *Poincaré disk* (the conformally equivalent rendering of the $\{\sigma > 0\}$ half-plane used in the proofs), drawn at the eikonal limit $e = 0$, where honesty (Postulate 1) makes the leaf exactly hyperbolic. The faint backdrop is the modular tiling: at $e = 0$ the constant-curvature metric makes its tiles *congruent* (equal cost-area) — a uniform cost grid; they crowd toward the boundary $\sigma=0$ (the Diracs) — the *wall* — which is thus infinitely many tiles away, at infinite distance. The highlighted geodesic joins two beliefs A and B ; honesty makes its beads *equal-cost steps* (the cheapest revision *opens up*; the straight chord, dashed, costs more). Along the *learning track* the posterior concentrates with evidence: the fixed beliefs labelled by evidence count $n = 1, 5, 20, 60$ rest ever nearer the wall — specifically the target Dirac δ_{x^*} , the certainty at the true state — their steps shrinking without ever reaching it (the standardized base then nearer Gaussian, remark 4.20). At $e = 0$ the curvature is constant, $K = -1/(2cJ_0)$ ($-1/(4J_0)$ at the representative $c=2$, a unit image, theorem 4.23); for $e > 0$ it varies (theorem 4.18). (b) The same leaf as a Gaussian *pseudosphere* (local model, Hilbert’s theorem (do Carmo, 1976)): the width equals the spread σ (iso-entropy circles), and the same fixed beliefs $n = 1, \dots, 60$ stand ever nearer the cusp ($H \rightarrow -\infty$): in both panels the disk boundary, the cusp and the *wall* are one and the same. Off-figure, among location-scale leaves the Gaussian ($J_0=1$, Stam) is the most curved (theorem 4.19). All quantities are verified numerically.

figure 2 reads this on a single location-scale leaf: panel (a) as the *Poincaré disk* (the conformal image of the $\{\sigma > 0\}$ half-plane) — the wall at $\sigma=0$, the cheapest revision *opening up* (widening the spread) — and panel (b) as a pseudosphere whose width is the spread σ .

We call these geometries *belief-cost geometry* and keep the term throughout. Two distinctions fix its place: here *belief* is a Bayesian density (a posterior over a fixed system), not the Dempster–Shafer belief function nor the decision-theoretic Bayes cost; and *cost* is the length of a belief revision, distinct from distinguishability, dissipation length, and free-energy-principle geometries (Section 7). A direct algorithmic reading — a Kalman filter revising at finite cost *vs* a network frozen on a certainty, which must be retrained — makes the wall concrete (box 1).

3 Setting

We first state the question, then build the framework in layers of decreasing necessity: the *object* (forced by finiteness), the two *postulates* (the posed and defended choices), and the remaining *assumptions* (transparent, a technical restriction, or borrowed). Table 1 gathers every item with its status and its entropic reading. Each statement below carries this status as a tag: (*setting*) for the modeling frame, *Postulate* for a posed choice, and **Bn** for a borrowed tool; the rest is proved here. Provenance is further labelled *tier-P* (physical/interpretive — motivates, never a proof),

tier-E (borrowed — a standard theorem cited), or *tier-D* (proved here, in pure metric geometry); the full dependency map is table 3.¹

3.1 Definitions and setting assumptions

An agent has no access to the state of the world x , assumed fixed: a sensor interposes itself, and the agent receives only observations y . Bayesian inference answers this, through Bayes' theorem (MacKay, 2003, p. 24). We adopt a static reading — a Bayesian posterior over the *fixed* state x (MacKay, 2003, p. 460), not a sampling-theory confidence interval (MacKay, 2003, p. 464): the state is fixed, the agent has a finite number of observations, and we ask what it may, at a given instant, legitimately believe.

We model the sensor as a finite-resolution, noisy channel, the transition matrix $Q_{j|i} = P(y = b_j | x = a_i)$ giving the probability of observing b_j when the world is in state a_i , with forward map $p_Y = Q p_X$ (MacKay, 2003, p. 146): the agent sees a Q -filtered version of the world. The output of inference is then a *belief*, the posterior $p(x | y) = p(y | x)p(x)/p(y)$ — *posterior = likelihood $\times$ prior / evidence* (MacKay, 2003, p. 29, also p. 50); inferring is the inversion of this known forward channel (inference, decoding and perception, one inverse problem). The belief is thus a *density* over the states of the world.

To this belief certainty is denied *twice*, and both denials diverge at the boundary where the Fisher information J blows up. First through observational finiteness. Specifying the world to a resolution δ costs $H(X) + \ln(1/\delta)$ nats, diverging as $\delta \rightarrow 0$ — the quantized differential entropy (Cover and Thomas, 2006, §8.3), whose discrete analogue is the essential bit content H_δ (MacKay, 2003, p. 75); the same barrier reads in estimation: by Cramér–Rao the variance of an unbiased estimator satisfies $\text{Var}(\hat{\theta}) \geq 1/J(\theta)$, a zero variance would require $J = \infty$ (Cover and Thomas, 2006, Th. 11.10.1); and since the channel capacity is finite, processing creates no information, $I(x; \hat{x}) \leq I(x; y)$ (data processing, Cover and Thomas, 2006, Th. 2.8.1) $\leq NC$ (Cover and Thomas, 2006, Ch. 7). Then through physical finiteness: erasing information dissipates at least $k_B T \ln 2$ per bit (Landauer, 1961; Bennett, 1982), verified experimentally (Bérut et al., 2012), and the information capacity of a region of bounded size and energy is itself bounded (Bekenstein, 1981). Reaching certainty would require infinite energy and memory. The density comes from inference; the two forms of finiteness only exclude its contraction to a Dirac. figure 1 summarizes this chain, from the noisy world to the belief-density.

The belief is the Bayes posterior $p(x | y_{1:n})$ — the coherent epistemic state of a finite agent (Cox, 1946; Jaynes, 2003), not an arbitrary density. Bayes thus *motivates* the object (the coherent epistemic state) and the twofold refusal of certainty just stated; it enters *none* of the proofs below, which take only that the belief is a regular density (assumption 1). Bayesian updating is moreover *multiplicative* — a vertical reweighting of the density, $p(x | y_{1:n+1}) \propto p(x | y_{1:n})p(y_{n+1} | x)$ —, whereas the cost geometry we build is a *horizontal* transport: the geometry *prices* a revision, it does not *enact* one. We study, at a fixed instant, the cost geometry of the posterior, not the dynamics of updating it.

Finiteness thus settles the *object* — a density, certainty excluded — but leaves the geometry open. To turn the excluded certainty into distances, a wall and a curvature, we model what it costs to *change* a belief — the subject of the two postulates below (Section 3.2). What the frame then *forces* — the wall, then the honest family — is the subject of Section 4.

Definition 3.1 (belief, entropy, Fisher; the carrier set). A *belief* is a density $p \in \mathcal{P}_2(\mathbb{R}^n)$, the probability measures of finite second moment $m_2 := \int |x|^2 p$; its entropy (Shannon, nats) is $H(p) = -\int p \ln p dx$, its knowledge $-H(p)$, its Fisher information $J(p) = \int |\nabla p|^2 / p \in [0, +\infty]$. We write $\text{Ent} := -H$ and $\partial_\infty := \{H = -\infty\}$ the *boundary of certainties* (singular measures, including all Diracs). The carrier $\mathcal{P}_2(\mathbb{R}^n)$ is forced by finiteness (the object); the *metric* that turns

¹The letter n carries two context-disambiguated meanings: the ambient state-space dimension ($\mathbb{R}^n$) and the evidence count ($n = 1, 5, \dots$, figure 1).

it into the cost *arena* — the Wasserstein distance $\mathcal{W}_2$ — is the first modeled choice (the metric, Postulate 0), its relief fixed by the second (honesty, Postulate 1), not posited here. Gaussian dictionary ($n = 1$): $H = \frac{1}{2} \ln(2\pi\sigma^2) + \frac{1}{2}$, $J = 1/\sigma^2$.

Definition 3.2 (regularity class (*setting*)). $\mathcal{R} \subset \mathcal{P}_2(\mathbb{R}^n)$: C^1 densities, strictly positive, with finite m_2 and J . It is the interior domain; ∂_∞ is excluded from it.

Assumption 1 (A1 — object (*setting*)). The belief is a density of class $\mathcal{R}$ (definition 3.2), a point of the arena. Finiteness *closes off the Dirac* (∂_∞ excluded: infinite precision would make J infinite there); the continuous C^1 class is a working idealization.

Remark 3.3 (two operational constraints (*setting*)). Two constraints drive the characterization. (a) *Boundary excluded*: ∂_∞ (infinite Fisher information, definition 3.1) is out of budget, the belief lives in the interior ($\mathcal{R}$). (b) *Well-posed*: inference — minimization over the arena — must admit an *interior* minimizer, with no collapse toward ∂_∞ . (*The boundary is costly because J diverges there; J is here the slope of $-H$, not the Fisher–Rao tensor — principle 1.*)

Definition 3.4 (plausibility and relief (*setting*)). The plausibility of a belief — no longer the posterior $p(x | y)$ over the states of the world, but a degree over the beliefs themselves — is modeled by a Gibbs measure on the space of beliefs (Jaynes, 2003),

$$\Pi \propto \nu \exp(-U), \quad U = -\ln \frac{d\Pi}{d\nu} \geq 0,$$

ν being a reference measure. The *relief* U is the negated log-density: an *improbability* (least commitment). The boundary of certainties ($-H = +\infty$) is all the less plausible the more the relief grows there; no form of U is fixed here.

Π is a measure on the space of beliefs: a density, not an evolution. The relief U remains *generic* — lower semi-continuous, ≥ 0 —, never a fixed functional form.

Assumption 2 (A2 — measure (*setting*)). Knowledge is measured by an *information* measure; among those satisfying continuity, maximality on the uniform, expansibility and grouping (recursivity), the Shannon entropy is the *unique* choice up to a unit (Shannon–Khinchin (Shannon, 1948; Khinchin, 1957)). Operationally $-H$ is the description cost the source-coding theorem makes inevitable (MacKay, 2003, thm 4.1, p. 78, with the asymptotic equipartition property (AEP) p. 80). This is a *discrete* characterization; for continuous beliefs it does not transfer literally (the differential entropy is neither nonnegative nor reparametrization-invariant), but it motivates reading knowledge on the differential entropy $H = -\int p \ln p$ of definition 3.1, knowledge being $-H$ — defined up to an additive constant (a reference scale) that drops out of every result, since these use only the entropy *slope* ∇H (this additive freedom is distinct from, and orthogonal to, the *multiplicative* cost gauge of theorem 4.23). The Shannon–Khinchin axioms make this the natural reading once one elects to measure knowledge by an information measure at all — itself a framing choice: the measure is thus *transparent and motivated* rather than merely imposed, though, unlike the object forced by finiteness, the axioms (notably grouping) remain a posed normative choice (relax grouping $\rightarrow$ Rényi). (*Independently and downstream: Shannon is the entropy for which the Otto identity assumption 4 makes the transport slope $\sqrt{J}$; this coupling to the arena is specific to it.*)

3.2 The two postulates

What is costly to the agent is to *change* belief. To turn the excluded certainty into distances we make *two* posed, defended commitments — the *arena* (Postulate 0) and *honesty* (Postulate 1), nested. Each is a choice an informed reader could refuse, and each is defended; together they pin the cost geometry, while every other ingredient is forced, transparent, or borrowed.

Postulate 0 (the cost is a scalar price on optimal transport). We model the *cost* of changing a belief as a *scalar price field over the optimal transport* of probability mass on the world’s state space, at fixed mass — a transport metric, reweighted by a scalar relief. Two clauses, both contestable: the cost is a *transport* (mass moved over the state space, not a reweighting in memory), and the price is *scalar* (isotropic — it depends on the belief, not on the direction of the revision). This is a statement about *cost*, not mechanism: a belief is *updated* by Bayes (a multiplicative reweighting), whereas the *price* of any revision is read as a transport length. The non-conformal (anisotropic) alternative is left open (remark 6.1).

From this commitment the arena *follows* rather than being posited. Mass conservation (a belief is a probability, total mass one), locality (the total cost is *additive* over a ground cost $c_0(x, y)$ on elementary state displacements — the Kantorovich form $\int c_0 d\pi$, excluding coupling functionals non-additive in the plan (Santambrogio, 2015; Villani, 2009)) and the demand for a length structure single out an optimal-transport metric; a power ground cost $d(x, y)^p$ then pins the Wasserstein family W_p . The quadratic case $p = 2$ is fixed by a single *intrinsic* requirement, prior to any result below: to speak of geodesics, a wall and a curvature *at all*, the length must be *Riemannian* — a differentiable kinetic action $\frac{1}{2} \int |\dot{\gamma}|^2$ (Benamou–Brenier) admitting the Maupertuis–Jacobi correspondence on which the machinery of Section 4 rests. Among the W_p , only W_2 is Riemannian (Otto, 2001; Ambrosio et al., 2008; Gigli, 2012); $W_{p \neq 2}$ is a metric but carries no such differential structure — standard optimal-transport folklore, with the necessity direction ($p \neq 2$) made exact on the location-scale leaf (a single location-scale family, a 2-D *leaf* of $\mathcal{P}_2$) in proposition 3.5 (failure already on a single 2D leaf rules out a Riemannian structure on the whole space; the positive global Riemannian structure of W_2 stays the borrowed fact above). This fixes $p = 2$ on its own terms — *before*, and independently of, the downstream payoffs it happens to unlock (the Otto slope $\|\nabla_{W_2} H\|^2 = J$, assumption 4; the flat location-scale base, assumptions 7 and 10). For a digital twin, whose role is to *represent* the world, we locate this cost in the *represented* space (the states), not in the machine’s memory — itself part of the same commitment, not a deduction from the twin’s role; the parameter-space reading (elastic weight consolidation, box 1) is the explicit alternative.

Proposition 3.5 (the quadratic exponent is forced). *Within the transport family, $p = 2$ is the unique exponent for which the cost geometry is Riemannian (and, heuristically, the one for which the slope of $-H$ is the classical L^2 Fisher information). On any location-scale family with standardized base φ_0 (quantile Q_0), the quantile isometry (assumption 10; the general- p W_p/L^p quantile identity, (Santambrogio, 2015, §2)) makes the W_p length element of a tangent ($d\mu, d\sigma$) the L^p norm of $d\mu \mathbf{1} + d\sigma Q_0$ on the plane $\text{span}\{\mathbf{1}, Q_0\} \subset L^p(0, 1)$ (well-defined when the base has a finite p -th moment, $Q_0 \in L^p$; automatic at $p = 2$, where finite variance suffices); by the Jordan–von Neumann theorem (Jordan and von Neumann, 1935) this norm derives from an inner product iff the parallelogram law holds on $\text{span}\{\mathbf{1}, Q_0\}$, which for any non-degenerate base (Q_0 not a.e. constant) occurs iff $p = 2$, where it equals $d\mu^2 + d\sigma^2$ — the flat half-plane (assumptions 7 and 10) — and is genuinely Finsler otherwise. On the W_2 arena the same exponent is what makes the tangent norm the L^2 kinetic energy $\int p |v|^2$, whose pairing with the L^2 score gives the Otto identity $\int p |\nabla \ln p|^2 = J$ (assumption 4, a separate global fact); heuristically, for $p \neq 2$ the corresponding slope of $-H$ is an L^q object ($\frac{1}{p} + \frac{1}{q} = 1$) rather than the classical L^2 Fisher. The Riemannian length required by the transport postulate (Postulate 0; Maupertuis) thus forces $p = 2$ (the parallelogram law on $\text{span}\{\mathbf{1}, Q_0\}$ holds iff $p = 2$; checked numerically for the Gaussian, Laplace, and a skewed base in the verification companion).*

This fixes the *exponent* within the transport family; that the cost is a transport *at all* remains Postulate 0, the one posed choice (the parameter-space alternative, box 1, is a Fisher–Rao cost on weights). It is orthogonal to remark 6.1. Dynamically the canonical gradient-flow structure of the heat equation lives in the same W_2 (the JKO theorem (Jordan et al., 1998): heat flow = W_2 gradient flow of Shannon entropy) — a tier-P corroboration that enters no proof (our reading is static).

Assumption 3 (**A3** — arena (consequence of Postulate 0)). The geometry deforms *transport*: arena $(\mathcal{P}_2, \mathcal{W}_2)$ — the two-step consequence of Postulate 0 (transport *type*; then $p = 2$ for a Riemannian length, proposition 3.5; cf. principle 1 and remark 7.1).

Physical corroboration (motivates, never a proof). The same arena is *also reached* from physics: driving a density from p_0 to p_1 in time T dissipates at least $\mathcal{W}_2^2(p_0, p_1)/T$ (Benamou and Brenier, 2000; Dechant and Sakurai, 2019), so the physical price of a revision is itself a $\mathcal{W}_2$ length (Section 7). This is *not* an independent derivation, and *no proof depends on the physical reading*: invoking the dissipation floor for a *belief* assumes the very identification of Postulate 0 (it relocates the contestable step, it does not remove it). It stays tier-P — it *motivates*, never proves (table 3).

The arena and its scalar price still leave the relief U free: *every* relief that dominates the Fisher information raises the wall (theorem 4.10), so cJ , cJ^2 , or $J + \text{const}$ would all serve — Fisher is not yet singled out. One further commitment singles it out.

Postulate 1 (entropic honesty). The price is *honest*: one nat of knowledge costs the same length everywhere — the metric slope of knowledge $-H$ is constant across the space of beliefs. This is a choice an agent could refuse (a distorted price would charge confidence unevenly); we adopt it as the calibration a digital twin owes its own uncertainty. By the characterization theorem 4.15 it singles out, *among continuous reliefs*, a *unique family* (up to the unit c), $U = cJ$ — the Fisher family —, and the wall then follows ($cJ \in \mathcal{C}$). Its denials are live: $U = J + \text{const}$ dominates but is not honest; $U = cJ^\alpha$, $\alpha > 1$, raises the wall but prices nats unevenly.

P1 *presupposes* P0: the constant-slope condition is read on the transport metric (the slope is $\sqrt{J}$ by the Otto identity assumption 4, which lives on the arena). The two postulates are a *ladder*, not two independent axioms — P0 builds the arena and its scalar price, P1 fixes that price. The honesty reading aligns with Landauer (each nat dissipates the same $k_B T$); like the dissipation floor, this is *defense*, *not proof* — it enters no theorem (theorem 4.15 uses only the eikonal condition as a hypothesis; proposition B.1).

Principle 1 (the demotion of J — the engine of the framework). *The whole construction turns on a single demotion. In information geometry J is the metric tensor (Fisher–Rao, Čencov) — the ruler of distinguishability. Here, by the Otto identity $J = \|\nabla_{\mathcal{W}_2} H\|^2$ (assumption 4), it is instead the squared slope of $-H$ on the transport arena. This demotion is what lets honesty select $U = cJ$ (theorem 4.15) rather than restate it, and it makes the forbidden object ($J = \infty$ at the Dirac) and the priced one ($U = cJ$) two readings of one divergence — the slope of $-H$ at the boundary. Met once here, it pays off repeatedly: in the eikonal (theorem 4.15) and the curvature that reads J_0 (the Fisher information of the standardized base, $J_0 := \int (\varphi'_0)^2 / \varphi_0$, = 1 for the Gaussian; theorem 4.19), off which the positioning against Čencov is then read (remark 7.1); the wall is the exception — it needs only assumption 5, not the demotion.*

3.3 The cost class

This conformal class is the stage on which every result below is played. Postulate 0 prices a revision by a *scalar* — one number at each belief — and a scalar multiplying the transport metric is exactly a *conformal* reweighting: $\tilde{g} = 2(e + U) g_{\mathcal{W}_2}$, with the *relief* U the local price of precision, $e \geq 0$ a baseline level, and the “2” the forced unit (proposition 3.5). The conformal form is therefore the geometric face of “scalar price”. The three results are then three statements about *which* reliefs U make this geometry well-posed; the candidate class $\mathcal{U}$ gathers all admissible reliefs.

Definition 3.6 (space of candidates (*setting*)). A *relief* is a function $U : \mathcal{P}_2(\mathbb{R}^n) \rightarrow [0, +\infty]$. The *space of candidates* is

$$\mathcal{U} := \{ U : \mathcal{P}_2(\mathbb{R}^n) \rightarrow [0, +\infty] \mid U \text{ is } \tau\text{-sequentially l.s.c.} \},$$

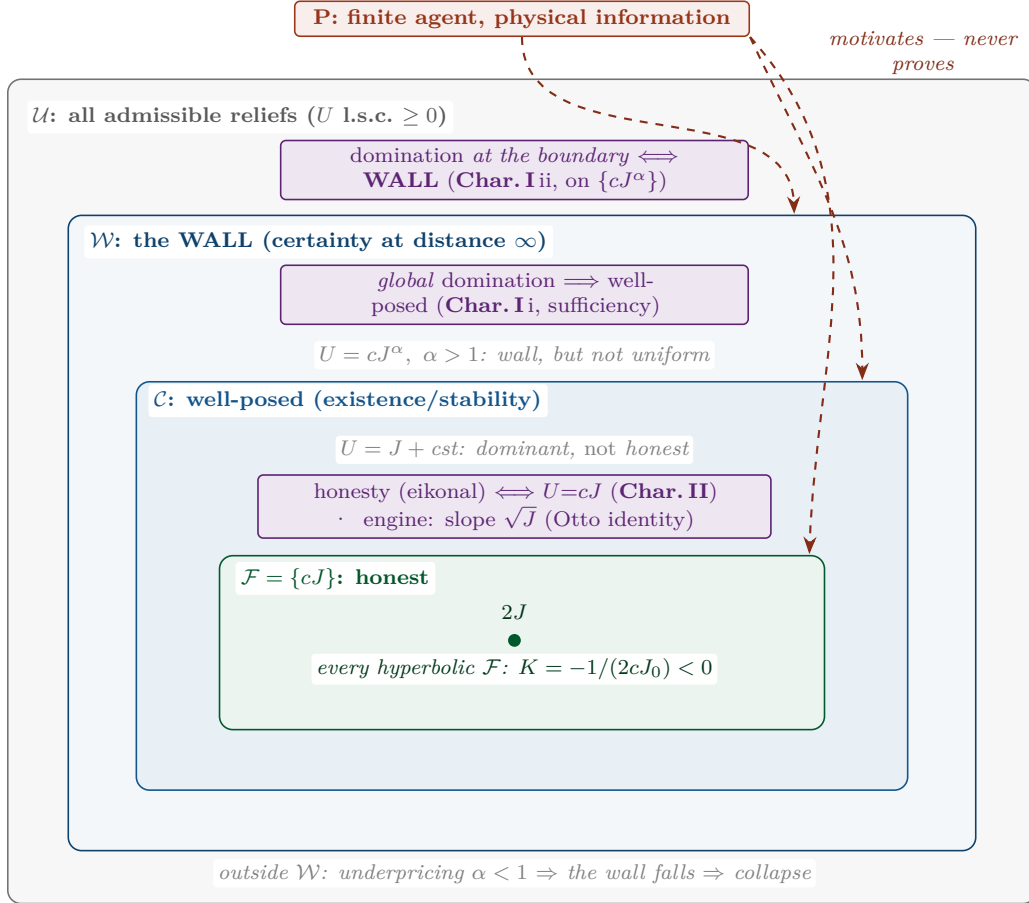


Figure 3: The universe of cost geometries. The nested rings $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$: the wall $\mathcal{W}$ (theorem 4.10), the well-posed class $\mathcal{C}$, the honest Fisher family $\mathcal{F}$ (theorem 4.15), and at their center the Gaussian, the most hyperbolic location-scale belief (Stam rigidity, theorem 4.19). The physics $\mathbf{P}$ *motivates* the boundaries (dashed) without entering any proof (proposition B.1).

τ being the narrow convergence. To each $U \in \mathcal{U}$ and each level $e \geq 0$ is associated the *cost geometry* (2: unit; $e > 0$ for licit reparametrization and separation of points, $e = 0$ for the eikonal/limiting case, conformal metric on $\{U > 0\}$)

$$\tilde{g}_{e,U} := 2(e + U)g_{\mathcal{W}_2}, \quad \ell_e(\gamma) := \int_0^T \sqrt{2(e + U(\gamma_t))} |\dot{\gamma}|(t) dt, \quad d_e := \inf \ell_e. \quad (3.1)$$

The candidate class is *conformal* — the scalar-price form posited in Postulate 0; non-conformal (anisotropic) deformations are the open horizon (remark 6.1).

The results then carve, inside $\mathcal{U}$, three nested subclasses $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$ (the Fisher family, the well-posed and the wall classes; defined below in definitions 4.8, 4.9 and 4.14; figure 3).

4 Characterizations

We first gather the borrowed tools (Section 4.1) and the machinery valid for every candidate (Section 4.2), then establish the three characterizations: the *boundary* (the wall), the *honest family* (the eikonal $U = cJ$), and the *curvature* (hyperbolicity and Stam rigidity).

The classes and the entropy reading used throughout are gathered at a glance in figure 3 and table 1.

Table 1: The framework and its geometry of $-H$ — one map. *Top:* every commitment with its status (exactly two posed postulates; all else forced, transparent, borrowed, or a unit). *Bottom:* each object re-read as a fact about the entropy $-H$ via the Otto identity — a presentation, adding no theorem.

<i>What makes $-H$ a geometry</i>		
Element	Commitment	Status
Object (assumption 1)	belief = density; certainty ($-H = +\infty$) excluded	forced (finiteness)
Measure (assumption 2)	knowledge = Shannon $-H$ (Shannon–Khinchin)	transparent
P0 (Postulate 0)	cost = scalar price on optimal transport $\mathcal{W}_2$	postulate
P1 (Postulate 1)	honest price — one nat, one length everywhere	postulate
Otto (assumption 4)	the slope of $-H$ is $\sqrt{J}$ — the hinge	borrowed
Cost class (definition 3.6)	conformal (scalar) relief $2(e + U)g_{\mathcal{W}_2}$	restriction (C1)
Exponent & unit ($p=2, c$)	$p=2$ forced by Riemannian length (proposition 3.5); c a cost gauge (theorem 4.23)	forced / unit
<i>Its geometry — each object read through $-H$</i>		
Geometric object	Entropic reading	Form
Cost floor	entropy-distance floor; $ \Delta H $ is only the entropy-axis projection (a pure location shift, $\Delta H = 0$, still costs)	$d_0 \geq \sqrt{2c} \Delta H $ (eq. along $\nabla(-H)$)
Fisher information	squared slope of $-H$ (Otto)	$J = \ \nabla_{\mathcal{W}_2} H\ ^2$
Honesty (eikonal)	constant entropy slope (κ , definition 4.13)	$\sqrt{J}/\sqrt{2U} = \kappa \Leftrightarrow U = cJ$
The wall (certainty)	divergence of the entropy	$\partial_\infty = \{H = -\infty\}$
Curvature (at $e = 0$)	inverse of the base entropy-slope ²	$K = -1/(2cJ_0)$
Stam rigidity (static)	at fixed variance: max entropy = min base slope = most-curved leaf	$J_0 \geq 1$, eq. iff Gaussian
Thermodynamic bridge	the floor is an energy (Landauer; tier-P)	$d_0 = (\sqrt{2c}/k_B T) E$

4.1 Borrowed tools

Assumption 4 (B1 — the Otto identity). On $(\mathcal{P}_2, \mathcal{W}_2)$, $-H$ is geodesically convex (McCann, 1997) and $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ — the Otto identity (Otto, 2001), here via Ambrosio et al. (2008, §10.4). *This is the crux of the argument, the converter “uniform price” $\rightarrow$ “cost $\propto J$ ”. (The equality is used only by the eikonal (theorem 4.15); the wall (theorem 4.10) needs only the upper-gradient inequality assumption 5.)*

Assumption 5 (B2 — strong upper gradient). For every AC (absolutely continuous) curve: $|\text{Ent}(\gamma_t) - \text{Ent}(\gamma_s)| \leq \int_s^t \sqrt{J} |\dot{\gamma}| dr$, the left-hand side finite and $\text{Ent} \circ \gamma$ absolutely continuous as soon as the right-hand side is (Ambrosio et al., 2008, §§1.2, 2.4).

Assumption 6 (B3 — compatibility $(\mathcal{W}_2, \tau)$). $(\mathcal{P}_2, \mathcal{W}_2)$ Polish and geodesic (Ambrosio et al., 2008, ch. 7); $\mathcal{W}_2$ is τ -l.s.c.; the $\{m_2 \leq M\}$ are τ -compact.

Assumption 7 (B4 — flat Gaussian leaf). $\mathcal{W}_2^2(\mathcal{N}(\mu_1, \sigma_1^2), \mathcal{N}(\mu_2, \sigma_2^2)) = (\mu_1 - \mu_2)^2 + (\sigma_1 - \sigma_2)^2$ (Givens and Shortt, 1984): the leaf $\mathcal{G} = \{\mathcal{N}(\mu, \sigma^2)\}$ is isometric to the Euclidean half-plane $\{\sigma > 0\}$; McCann’s geodesics stay Gaussian there (Ambrosio et al., 2008, ex. 7.3.14).

Assumption 8 (B5 — metric Hopf–Rinow). A locally compact length space in which bounded closed balls are compact is complete and geodesic (Burago et al., 2001, thm 2.5.28).

Assumption 9 (B6 — l.s.c. dual of Fisher). $J(q) = \sup_{\psi \in C_c^\infty} \{-2 \int \Delta \psi dq - \int |\nabla \psi|^2 dq\}$ for every $q \in \mathcal{P}(\mathbb{R}^n)$ (Dupuis and Ellis, 1997, lem. D.45); as a supremum of narrowly continuous functionals, J is narrowly l.s.c. (Hence $J \in \mathcal{U}$ and, for every $c > 0$, $cJ \in \mathcal{U}$.)

Assumption 10 (B7 — flatness of 1D location-scale families). The quantile map $p \mapsto F_p^{-1}$ is an isometry of $(\mathcal{P}_2(\mathbb{R}), \mathcal{W}_2)$ onto the convex cone of inverse cumulative distribution functions in $L^2(0, 1)$ (Santambrogio, 2015; Bobkov and Ledoux, 2019). For a base φ_0 (density, mean 0, variance 1, quantile Q_0), the location-scale family $\mathcal{L}_{\varphi_0} = \{p_{\mu, \sigma} = \sigma^{-1}\varphi_0((\cdot - \mu)/\sigma)\}$ is there the affine 2-plane $\{\mu \mathbf{1} + \sigma Q_0\}$, with $\langle \mathbf{1}, \mathbf{1} \rangle = 1$, $\langle \mathbf{1}, Q_0 \rangle = 0$, $\langle Q_0, Q_0 \rangle = 1$: $(\mathcal{L}_{\varphi_0}, \mathcal{W}_2)$ is thus *isometric to the Euclidean half-plane* $\{\sigma > 0\}$, with metric $d\mu^2 + d\sigma^2$. (Generalizes assumption 7: the Gaussian is the case $\varphi_0 = \mathcal{N}(0, 1)$.)

Metric setting. $\gamma : [0, T] \rightarrow \mathcal{P}_2$ is AC^2 if there exists $m \in L^2$ with $\mathcal{W}_2(\gamma_s, \gamma_t) \leq \int_s^t m$; $|\dot{\gamma}|$ exists a.e. (Ambrosio et al., 2008, thm 1.1.2). We set $\mathcal{E}(\gamma) := \sup_{\text{part.}} \sum_i \mathcal{W}_2^2(\gamma_{t_i}, \gamma_{t_{i+1}})/(t_{i+1} - t_i)$, the curve's metric *energy*.

Lemma 4.1 (kinetic action (metric energy)). (a) $\gamma \in AC^2 \Rightarrow \mathcal{E}(\gamma) = \int_0^T |\dot{\gamma}|^2$. (b) $\mathcal{E}(\gamma) < \infty \Rightarrow \gamma \in AC^2$. (c) $\mathcal{E}$ is τ -pointwise l.s.c.

Proof. (a) ($\leq$, i.e. $\mathcal{E} \leq \int |\dot{\gamma}|^2$) Cauchy–Schwarz on each subinterval, $\mathcal{W}_2^2(\gamma_{t_i}, \gamma_{t_{i+1}}) \leq (t_{i+1} - t_i) \int_{t_i}^{t_{i+1}} |\dot{\gamma}|^2$, sum and then take the sup. ($\geq$, i.e. $\int |\dot{\gamma}|^2 \leq \mathcal{E}$) the dyadic quotients $f_N \rightarrow |\dot{\gamma}|$ a.e. (a.e. limit, (Ambrosio et al., 2008, thm 1.1.2), not the Cauchy–Schwarz bound); Fatou: $\int |\dot{\gamma}|^2 \leq \liminf \int f_N^2 \leq \mathcal{E}$. (b) the inequality $\frac{(x+y)^2}{\alpha+\beta} \leq \frac{x^2}{\alpha} + \frac{y^2}{\beta}$ yields additivity of $\mathcal{E}$, hence absolute continuity and then $\int |\dot{\gamma}|^2 \leq \mathcal{E}$. (c) each term is τ -l.s.c. (assumption 6); a sup of l.s.c. is l.s.c. $\square$

Lemma 4.2 (entropy, slope — recall of imports). With H, J in the sense of definition 3.1: Ent $> -\infty$ on $\mathcal{P}_2$, geodesically convex (McCann, 1997); its metric slope equals $\sqrt{J}$ (assumption 4); it is a strong upper gradient (assumption 5).

4.2 What holds for every candidate: existence and geodesics

This entire section holds for *every* $U \in \mathcal{U}$: it is the common machinery of the whole class, independent of the choice of relief. For $\gamma \in AC^2$ joining p_0 to p_1 : $\mathcal{A}_T(\gamma) := \int_0^T [\frac{1}{2}|\dot{\gamma}|^2 + U(\gamma_t)]dt$, $\Phi(T) := \inf \mathcal{A}_T$, $\Psi(e) := \inf \ell_e = d_e$. The statements below hold for every candidate; their proofs, with their regime clauses, are collected in Appendix A.

Lemma 4.3 (AM–GM). $\forall a, b \geq 0$: $\frac{1}{2}a^2 + b \geq \sqrt{2b}a$, with equality iff $a = \sqrt{2b}$.

This inequality converts the action $\frac{1}{2}|\dot{\gamma}|^2 + (e + U)$ into length $\sqrt{2(e + U)}|\dot{\gamma}|$, with saturation at the Jacobi speed.

Theorem 4.4 (Maupertuis–Jacobi correspondence (Arnold, 1989)). $\forall U \in \mathcal{U} \forall e > 0$, writing $p_0, p_1 \in \{U < \infty\}$: (i) $\forall \gamma \in AC^2$: $\mathcal{A}_T(\gamma) \geq \ell_e(\gamma) - eT \geq \Psi(e) - eT$; (ii) if $\hat{\gamma}$ minimizes ℓ_e ($\Psi(e) < \infty$, $U(\hat{\gamma}) < \infty$ a.e.), its constant-energy reparametrization $|\dot{\hat{\gamma}}| = \sqrt{2(e + U)}$ exists, with parameter $T_e = \int |\dot{\hat{\gamma}}|/\sqrt{2(e + U)}$, and minimizes $\mathcal{A}_{T_e}$ with $\Phi(T_e) = \Psi(e) - eT_e$ (the bound $L/\sqrt{2e}$ requires $e > 0$: this is what makes the reparametrization licit); (iii) if γ^* minimizes $\mathcal{A}_T$ and saturates (i), then $\frac{1}{2}|\dot{\gamma}^*|^2 = e + U(\gamma^*)$ a.e.

Theorem 4.5 (existence of geodesics). $\forall U \in \mathcal{U}$, if $\Phi(T) < \infty$, then $\mathcal{A}_T$ admits a minimizer in $AC^2([0, T]; \mathcal{P}_2)$ joining p_0 to p_1 . (Same for ℓ_e , $e > 0$, (Buttazzo et al., 1998, §4.3).)

Proposition 4.6 (energy law). $\forall U \in \mathcal{U}$, every minimizer γ^* of $\mathcal{A}_T$ (provided by theorem 4.5, $\mathcal{A}_T(\gamma^*) < \infty$) satisfies $\frac{1}{2}|\dot{\gamma}^*|^2 = U(\gamma^*) + e$ a.e., for some constant $e \in \mathbb{R}$.

The constant $e \in \mathbb{R}$ thus obtained is, when it is > 0 , the *level* of definition 3.6 (which requires $e \geq 0$); its useful regime $e > 0$ is discussed in theorem 4.7.

Theorem 4.7 (optimal curve $\Leftrightarrow$ geodesic). $\forall U \in \mathcal{U}$, let γ^* be a minimizer of $\mathcal{A}_T$ and $e \in \mathbb{R}$ its constant (proposition 4.6); in the regime $e > 0$ (typically T small enough: by proposition 4.6 $-e \in \partial V(T)$ with $V(T)$ the reparametrization value function (Appendix A) convex, so $e(T)$ is

non-increasing, $e(T) \rightarrow +\infty$ as $T \rightarrow 0$ by kinetic dominance, and $e > 0$ may fail for large T — cusps), the only regime where ℓ_e is a distance: γ^* traverses its trace at the Jacobi speed, and under exact duality $\Phi(T) = \sup_{x>0} \{\Psi(x) - xT\}$ its trace is a geodesic of $\tilde{g}_{e,U}$ in the length sense (of $d_e = \inf \ell_e$; $\tilde{g}$ is not a smooth Riemannian metric for U merely l.s.c.). The converse is unconditional. (Exact duality is not automatic: it can fail through a first-order transition — a two-route counterexample (one short but costly, the other long but free) exhibits a jump in the energy.)

4.3 Characterization I: the wall

We first characterize the wall. Global domination of the Fisher information is *sufficient* for pushing certainty to infinite distance (every dimension); on the power family $\{cJ^\alpha\}$ boundary domination is moreover *necessary*; the general necessity is Conjecture C1 (definition 4.8 and remark 6.1). The map of admissible classes (figure 3) locates this wall: the ring $\mathcal{W}$ of reliefs that raise the boundary to infinite distance.

First equivalence. The operational constraint of remark 3.3(b) entails that the relief *dominates* the Fisher information *at the boundary*: on the power family, this is the *necessary* condition for the wall (theorem 4.10(ii)). *Global* domination, strictly stronger, is its unconditional *sufficient* form (theorem 4.10(i)); it defines the well-posed class (definition 4.8).

Definition 4.8 (well-posed class).

$$\mathcal{C} := \{U \in \mathcal{U} \mid \exists \varepsilon > 0, U \geq \varepsilon J \text{ pointwise on } \mathcal{P}_2\} \subset \mathcal{U}$$

(global domination: one same ε at every point).

Definition 4.9 (wall class). $\mathcal{W} := \{U \in \mathcal{U} : \partial_\infty \text{ is at infinite } d_e\text{-distance}\}$ — the reliefs that *raise the wall*. Global domination entails it (theorem 4.10(i)), hence $\mathcal{C} \subset \mathcal{W}$, strictly (on the family, $U = cJ^\alpha$ with $\alpha \in (1, \infty)$ raises the wall without belonging to $\mathcal{C}$). Whence the chain $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$.

Theorem 4.10 (characterization of the wall). (i) Sufficiency (every $U \in \mathcal{C}$: the wall).

$$U \geq \varepsilon J \implies \forall e > 0 \forall \gamma \in AC^2 : \ell_e(\gamma) \geq \sqrt{2\varepsilon} |H(p_1) - H(p_0)|;$$

— $-H$ is thus $\frac{1}{\sqrt{2\varepsilon}}$ -Lipschitz for d_e , and the wall holds: $\partial_\infty = \{H = -\infty\}$ is at infinite d_e -distance (every dimension), whereas it is at finite $\mathcal{W}_2$ distance. Since the wall pushes ∂_∞ to infinity, no minimizing sequence of finite cost collapses onto it: the minimizer of the action (theorem 4.5) is interior — with fixed endpoints, or for estimation as soon as the fidelity is coercive (the coercive-fidelity case is discharged in corollary 5.1); the general free-boundary case (mass escape) falls under remark 6.3. (ii) Necessity, on the family $U = cJ^\alpha$ ($c > 0, \alpha \in \mathbb{R}$). On this family (the equivalence witnessed on the Gaussian leaf; the forward direction $\alpha \geq 1 \Rightarrow$ wall holds in every dimension, the reverse $\alpha < 1 \Rightarrow$ no wall is read on the Gaussian leaf),

$$[\text{wall: } \partial_\infty \text{ at infinite } d_e\text{-distance}] \iff \alpha \geq 1 \quad (\text{domination at the boundary } J \rightarrow \infty).$$

$\alpha < 1$ brings ∂_∞ back to finite distance. (Global vs boundary domination.) Global domination (membership in $\mathcal{C}$) is strictly stronger: on this family, $cJ^{\alpha-1} \geq \varepsilon$ for every $J \in (0, \infty)$ forces $\alpha = 1$. Thus $\alpha \in (1, \infty)$ has the wall (boundary domination) without belonging to $\mathcal{C}$, which on the family coincides with $\alpha = 1$ (theorem 4.15). (iii) Scope. Sufficiency (i) is unconditional (every $U \in \mathcal{C}$); the general necessity (every $U \in \mathcal{U}$: “no boundary domination $\Rightarrow$ no wall”) is conjectured (remark 6.1).

Proof. (i) $U \geq \varepsilon J \Rightarrow 2(e + U) \geq 2\varepsilon J$, whence $\sqrt{2(e + U)} \geq \sqrt{2\varepsilon}\sqrt{J}$; since $\sqrt{J}$ is a strong upper gradient of $-H$ (assumption 5), $\ell_e \geq \sqrt{2\varepsilon} \int \sqrt{J} |\dot{\gamma}| \geq \sqrt{2\varepsilon} |\Delta \text{Ent}|$. If $\text{Ent}(p_1) = +\infty$, a finite joining

length is impossible (otherwise the strong upper gradient would make $\text{Ent} \circ \gamma$ a.c., hence finite at $t = 1$; or, by narrow l.s.c. of Ent , would force $\text{Ent}(p_1) \leq \liminf < \infty$): $\ell_e = +\infty$. Interiority of the action minimizer follows from theorem 4.5. (ii) On the Gaussian leaf $\mathcal{G}$ (assumption 7), the Fisher information of $\mathcal{N}(\mu, \sigma^2)$ is $J = 1/\sigma^2$ (elementary; the general scaling $J = J_0/\sigma^2$ is the computation of theorem 4.19(i), here $J_0 = 1$), and $U = c\sigma^{-2\alpha}$. Along the ray $\mu = \text{const}$, $\ell_e = \int \sqrt{2(e+U)} d\sigma$ *exactly* (witness path, for *convergence*); and $|\dot{\gamma}| \geq |\dot{\sigma}|$ gives, for any curve, the *lower bound* $\ell_e \geq \int \sqrt{2(e+U)} |\dot{\sigma}|$. At $e = 0$, near $\sigma = 0$, $\sqrt{2U} d\sigma = \sqrt{2c} \sigma^{-\alpha} d\sigma$ and $\int_0^{\sigma_0} \sigma^{-\alpha} d\sigma < \infty \Leftrightarrow \alpha < 1$: thus $\alpha \geq 1$ makes the radial lower bound diverge (wall) and $\alpha < 1$ gives the witness path a finite length (boundary at finite distance). The case $e > 0$: $\sqrt{2U} \leq \sqrt{2(e+U)} \leq \sqrt{2e} + \sqrt{2U}$ preserves divergence ($\alpha \geq 1$, via the lower bound $\sqrt{2U}$) and convergence ($\alpha < 1$, via the upper bound $\sqrt{2e} + \sqrt{2U}$, integrable on $(0, \sigma_0]$). The $\alpha \geq 1$ wall is moreover *global*, not merely radial: on the boundary tail $\{J \geq 1\}$, $\sqrt{2(e+U)} \geq \sqrt{2c} J^{\alpha/2} \geq \sqrt{2c} \sqrt{J}$, so any AC curve reaching ∂_∞ has $\ell_e \geq \sqrt{2c} \int_{\{J \geq 1\}} \sqrt{J} |\dot{\gamma}| = +\infty$ by assumption 5 (part (i) localized to the boundary, $\varepsilon = c$, valid off the leaf): the complement contributes $\int_{\{J < 1\}} \sqrt{J} |\dot{\gamma}| \leq \int_0^T |\dot{\gamma}| dt < \infty$ (any AC curve has finite $\mathcal{W}_2$ -length), so the divergence of $\int_0^T \sqrt{J} |\dot{\gamma}|$ forced by $\text{Ent} \rightarrow +\infty$ through assumption 5 localizes to $\{J \geq 1\}$; whence $cJ^\alpha \in \mathcal{W}$ for $\alpha \geq 1$, hence $\mathcal{C} \subsetneq \mathcal{W}$. (iii) cf. remark 6.1. $\square$

Corollary 4.11 (the wall, answer to the problem of finiteness). *Under remark 3.3 (boundary excluded + tenable exit), a conformal cost geometry is well-posed for existence and stability only if it raises the wall, hence only if its relief dominates J at the boundary (on the power family $\{cJ^\alpha\}$, $\alpha \geq 1$; theorem 4.10(ii)). On the power family, the wall is the condition that finiteness requires (open boundary excluded $\rightarrow$ collapse). The class $\mathcal{C}$ (global domination) is its convenient uniform form; on the family it restricts to $\alpha = 1$, in agreement with theorem 4.15.*

Remark 4.12 (the wall is a discriminating condition — the Dirac is *finite* in the base). In the transport base the Dirac sits at *finite*, reachable distance — $\mathcal{W}_2(\mathcal{N}(\mu, \sigma^2), \delta_\mu) = \sigma$ (assumption 7) —, so the relief must perform positive work, a divergent radial cost $\int \sqrt{2U} d\sigma$, to push certainty to infinity. The wall is therefore a *discriminating* condition on U : on the power family $\{cJ^\alpha\}$ it is exactly the dichotomy $\alpha \geq 1$ vs $\alpha < 1$ (theorem 4.10(ii)), whether it holds for *every* admissible relief being Conjecture C1 (remark 6.1). What the wall needs is domination *at the boundary*; the *global* domination of the class $\mathcal{C}$ (definition 4.8) is the stronger, sufficient form. None of this has an analogue in Fisher–Rao, whose boundary already lies at infinite distance *for free* ($ds^2 = (d\mu^2 + 2d\sigma^2)/\sigma^2$ diverges radially whatever the relief): there the wall discriminates nothing (table 2).

4.4 Characterization II: eikonal $\Leftrightarrow$ Fisher family

This second equivalence is the *characterization of Postulate 1* (Postulate 1): we make precise that demanding an honest price — the same length per nat at every point — selects exactly the reliefs proportional to Fisher information, the family $\mathcal{F} = \{cJ\}$. The selection is possible only because of the demotion (principle 1): J enters as the *slope* of $-H$, not as the ambient ruler — so honesty imposes a real condition (in a distinguishability geometry, where J is the ruler, there would be nothing to select). It uses the scalar (conformal) form of Postulate 0 *essentially*: the constant-slope condition reduces to $U \propto J$ only because the reweighting is scalar; for an anisotropic deformation it would not separate (remark 6.1). This is a uniqueness of *family*, not of object.

Definition 4.13 (honesty of the price — eikonal). A relief U , continuous and positive on $\{0 < U < \infty\}$, is *eikonal* if knowledge $-H$ has a constant metric slope for $\tilde{g}_{0,U}$: $\exists \kappa > 0$ such that $\|\nabla_{\tilde{g}_{0,U}} H\| = \kappa$ on $\{0 < U < \infty\}$ (where $\|\nabla_{\mathcal{W}_2} H\| = \sqrt{J}$, assumption 4).

Definition 4.14 (Fisher family). $\mathcal{F} := \{U = cJ : c > 0\} \subset \mathcal{C}$: the reliefs proportional to Fisher information (exponent $\alpha = 1$; $\mathcal{F} \subset \mathcal{C}$ by proposition 4.25).

Theorem 4.15 (characterization of the eikonal family). *For every relief U continuous and positive on $\{0 < U < \infty\}$ (in particular every member of $\mathcal{F}$ restricted to a location-scale leaf, where J is smooth; on all of $\mathcal{P}_2$ only the l.s.c. of assumption 9 is imported):*

$$U \text{ eikonal} \iff \exists c > 0, U = cJ \text{ pointwise on } \{0 < U < \infty\},$$

and then $\kappa = 1/\sqrt{2c}$. Eikonality of the price thus selects exactly the Fisher family $\mathcal{F} = \{cJ : c > 0\}$ (definition 4.14) on the effective domain $\{0 < U < \infty\}$: the equivalence pins U to cJ wherever the price is active, the eikonal-irrelevant boundary sets $\{U = 0\}$ and $\{U = \infty\}$ being left free. (Corollary: on power laws $U = cJ^\alpha$, eikonal $\Leftrightarrow \alpha = 1$.)

Proof. Conformal contraction (assumption 4): on $\{0 < U < \infty\}$, where U is continuous and positive, the metric slope rescales by the local conformal factor — $d_0(x, y) \sim \sqrt{2U(x)} \mathcal{W}_2(x, y)$ as $y \rightarrow x$ —, so for $\tilde{g}_{0,U} = 2U g_{\mathcal{W}_2}$ and $\|\nabla_{\mathcal{W}_2} H\| = \sqrt{J}$ we have $\|\nabla_{\tilde{g}_{0,U}} H\| = \sqrt{J}/\sqrt{2U}$ (continuity is what licenses this pointwise rescaling; for a merely l.s.c. U it can fail at a jump). This slope equals a constant $\kappa > 0$ at every point if and only if $\sqrt{J}/\sqrt{2U} = \kappa$, i.e. $U = J/(2\kappa^2) = cJ$ with $c = 1/(2\kappa^2)$; conversely $U = cJ$ gives $\kappa = 1/\sqrt{2c}$. $\square$

Remark 4.16 (uniqueness of family, not of object). theorem 4.15 is a *uniqueness of proportionality* ($U \propto J$, family $\mathcal{F} = \{cJ\}$), not of the constant: the *whole* family is eikonal, its members differing only by the unit $\kappa = 1/\sqrt{2c}$ (the “price of a nat”). We thus claim no unique object; we have characterized a *family*.

Remark 4.17 (honesty of the price). The eikonal condition is an honesty of pricing: each nat costs the same length everywhere, and (via Landauer (Landauer, 1961; Bérut et al., 2012)) the same energy $k_B T$ — one non-distortion in two units. The bridge is the Otto identity $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ (assumption 4), turning “uniform price” into proportionality to J . It reads off an already-established geometry (theorem 4.15); it enters no proof, all theorems being in nats. At the boundary distance and energy diverge together: at $e = 0$, $d_0 \geq \sqrt{2c} |\Delta H| = (\sqrt{2c}/k_B T) E$ (corollary 5.3), with equality along $\nabla(-H)$.

4.5 Characterization III: Hyperbolicity and Stam rigidity

The eikonal cost geometry is hyperbolic, and its curvature reads the Fisher information of the base — the demotion (principle 1) paying once more: what was the slope of $-H$ now surfaces as curvature. On every location-scale family, at $e = 0$, it equals (constant) $K = -1/(2cJ_0)$ (for $e > 0$ the curvature varies, theorem 4.18), and the Stam bound makes the Gaussian extremal — the most hyperbolic location-scale belief. What is unit-invariant — the *invariant content* — is the sign, the *ranking* of the leaves by curvature and its extremum (the Gaussian); the value $-\frac{1}{4}$ is its image under the unit $c = 2$ (remark 4.22).²

Theorem 4.18 (universal hyperbolicity of $\mathcal{F}$). $\forall U = cJ \in \mathcal{F} \forall e \geq 0$, on the Gaussian leaf $(\mathcal{G}, \tilde{g}_{e,U})$, the Gauss curvature equals, in closed form,

$$K(\mu, \sigma) = -\frac{c(c + 3e\sigma^2)}{2(c + e\sigma^2)^3} < 0 \quad (\forall \sigma > 0).$$

The whole Fisher family is thus strictly hyperbolic. At $e = 0$: $K \equiv -1/(2c)$ for all σ (exact Poincaré half-plane (do Carmo, 1976)). For $e > 0$: K increases from $-1/(2c)$ ($\sigma \rightarrow 0$) to 0 ($\sigma \rightarrow \infty$) — monotone, $\partial_\sigma K = 6ce^2\sigma^3/(c + e\sigma^2)^4 > 0$. The ranking of the leaves by curvature is defined at $e = 0$, where K is constant; this is the setting of Stam rigidity (theorem 4.19) and of theorem 4.23.

²The two closed forms ($K < 0$, and $K = -1/(2cJ_0)$ on the location-scale families) are confirmed symbolically (sympy); scripts and outputs in the verification companion (the `companion/` directory; run `make verify`, `deps numpy/scipy/matplotlib/sympy`). A public archive (repository snapshot with DOI) will be linked in the camera-ready.

Proof. $\tilde{g}_{e,U} = f(d\mu^2 + d\sigma^2)$, $f = 2e + 2c/\sigma^2$; for a conformal metric of the plane, $K = -\Delta(\ln f)/(2f)$ (Δ Euclidean, f depending only on σ). This classical formula is licit even though U is merely l.s.c. on $\mathcal{P}_2$ (theorem 4.7): restricted to the leaf, $f = 2e + 2c/\sigma^2$ is smooth on $\{\sigma > 0\}$. Direct computation:

$$\Delta(\ln f) = \partial_\sigma^2(\ln f) = \frac{2c(c + 3e\sigma^2)}{\sigma^2(c + e\sigma^2)^2}, \quad 2f = \frac{4(c + e\sigma^2)}{\sigma^2},$$

whence $K = -\frac{2c(c + 3e\sigma^2)/(\sigma^2(c + e\sigma^2)^2)}{4(c + e\sigma^2)/\sigma^2} = -\frac{c(c + 3e\sigma^2)}{2(c + e\sigma^2)^3}$. Numerator and denominator strictly positive ($c > 0$, $e \geq 0$, $\sigma > 0$): $K < 0$ everywhere. The limits and monotonicity follow by direct reading. $\square$

Theorem 4.19 (Stam rigidity — the Gaussian, the most hyperbolic location-scale belief). *Let φ_0 be a standardized base — absolutely continuous and strictly positive, with $\int x\varphi_0 = 0$, $\int x^2\varphi_0 = 1$ and finite Fisher information $J_0 := \int (\varphi'_0)^2/\varphi_0 < \infty$ (φ'_0 the a.e. derivative; C^1 is not required for the leaf curvature — weaker than the belief class $\mathcal{R}$ of definition 3.2 —, so e.g. the Laplace base ($J_0 = 2$) is covered) and $\mathcal{L}_{\varphi_0}$ the location-scale family it generates (assumption 10). For the representative $U = 2J$, at $e = 0$:*

- (i) $(\mathcal{L}_{\varphi_0}, \mathcal{W}_2)$ is isometric to the Euclidean half-plane (assumption 10), and $J(p_{\mu,\sigma}) = J_0/\sigma^2$, whence $U = 2J_0/\sigma^2$;
- (ii) the intrinsic Gauss curvature of $(\mathcal{L}_{\varphi_0}, \tilde{g}_{0,2J})$ is constant,

$$K_{\varphi_0} = -\frac{1}{4J_0} < 0;$$

- (iii) (Stam extremum) $J_0 \geq 1$ (Gaussian extremality of information at fixed variance, broad sense; cf. Stam, 1959; Cover and Thomas, 2006, Ch. 17), with equality if and only if φ_0 is Gaussian (assumption 9, $\text{Var} = 1$); hence $K_{\varphi_0} \in [-\frac{1}{4}, 0)$, and the extremal value $-\frac{1}{4}$ is attained by the Gaussian leaf (gauge $c = 2$, $\text{Var}(\varphi_0) = 1$; remark 4.24). (Scope outside location-scale: remark 6.2.)

Proof. (i) Flatness: assumption 10. Scaling: $J(p_{\mu,\sigma}) = \int (p'_{\mu,\sigma})^2/p_{\mu,\sigma} = \sigma^{-2} \int (\varphi'_0)^2/\varphi_0 = J_0/\sigma^2$. (ii) On $\mathcal{L}_{\varphi_0}$, $g_{\mathcal{W}_2} = d\mu^2 + d\sigma^2$ (assumption 10) and $\tilde{g}_{0,2J} = f(d\mu^2 + d\sigma^2)$ with $f = 2U = 4J_0/\sigma^2$; $\Delta(\ln f) = 2/\sigma^2$, $2f = 8J_0/\sigma^2$, whence $K = -(2/\sigma^2)/(8J_0/\sigma^2) = -1/(4J_0)$. (iii) Cramér–Rao/Stam for a density of variance 1 (location Fisher information J_0): $J_0 \text{Var} = J_0 \geq 1$, with equality iff the score φ'_0/φ_0 is affine. For an absolutely continuous φ_0 of finite Fisher information, the Cauchy–Schwarz equality forces φ'_0/φ_0 affine a.e.; the mean-zero normalization $\int x\varphi_0 = 0$ kills the additive constant, leaving $\varphi'_0/\varphi_0 = -kx$, which integrates to $\varphi_0 \propto \exp(-kx^2/2)$: φ_0 is Gaussian. (Laplace gives strict inequality: its a.e. score $-\sqrt{2} \text{sgn}(x)$ is not globally affine, whence $J_0 = 2 > 1$.) Whence $-1/(4J_0) \in [-\frac{1}{4}, 0)$, extremal at the Gaussian. (“Stam” in the broad sense: the bound used is the Gaussian extremality of information under fixed variance.) $\square$

Remark 4.20 (the Gaussian as the perfectly-informed case-limit). We read Stam rigidity (theorem 4.19) statically. The gap

$$J_0 - 1 \geq 0, \quad J_0 - 1 = 0 \iff \varphi_0 \text{ Gaussian},$$

is a *gauge-invariant index of non-Gaussianity* (the ratio $K(\text{gauss})/K(\varphi_0) = J_0$, remark 4.24). We compare beliefs as *frozen* objects: the posterior is indexed by the *quantity* of evidence n , a label, not a time; a belief resting on more observations *is* more Gaussian, it does not evolve towards one. Two borrowed facts place the Gaussian ($J_0 = 1$) as the perfectly-informed case-limit: *exactly* in the linear-Gaussian (Kalman) regime (box 1), and *asymptotically* under regularity (Bernstein–von Mises, the index $J_0 - 1 \rightarrow 0$ (van der Vaart, 1998); misspecification or outliers fall outside this regime (Kleijn and van der Vaart, 2012), cf. the M-estimator footnote in Section 7). The Gaussian is thus the canonical, perfectly-informed case-limit of the family, while every φ_0 with $J_0 > 1$ is a legitimate belief and $K_{\varphi_0} = -1/(2cJ_0)$ stays the principal object.

Remark 4.21 (why *position* information, and not scale). What distinguishes this rigidity is the *choice of the quantity* that drives the curvature. The conformal cost charges the *slope* of $-H$ along transport (the Otto identity, $\|\nabla_{\mathcal{W}_2} H\|^2 = J$, assumption 4), and J is the *position* Fisher information: $J(p_{\mu,\sigma}) = J_0/\sigma^2$ with $J_0 = \int (\varphi'_0)^2/\varphi_0$. This is the quantity that Cramér–Rao/Stam bounds below ($J_0 \geq 1$, variance 1). The Fisher–Rao geometry of location-scale families (symmetric base) is also hyperbolic of negative curvature, but governed by the *scale* information (Atkinson and Mitchell, 1981; Costa et al., 2015) — which Stam does not bound: hence the *absence* of this extremum in Fisher–Rao. *The conformal cost of transport makes position information carry the curvature, which is what makes Stam rigidity possible* (remark 7.1). The imputation is legitimate precisely because the transport base is *flat* (assumption 10): on a location-scale family all curvature comes from the Fisher relief, none from the base — whereas Fisher–Rao, whose base is curved, cannot cleanly separate tensor from relief. Moreover, $K = -1/(4J_0)$ holds for *every* standardized base — assumption 10 gives $g_{\mathcal{W}_2}$ flat without a symmetry hypothesis —, whereas the Fisher–Rao comparison requires a symmetric base: cost rigidity thus covers a *larger* class of bases.

Remark 4.22 (the honest representative $U = 2J$). Within $\mathcal{F}$ we *choose* the representative $c = 2$,

$$\boxed{U = 2J}, \quad \tilde{g}_{e,2J} = 2(e + 2J) g_{\mathcal{W}_2},$$

a *unit normalization* (one nat = two lengths: at $e = 0$, $\kappa = 1/2$, $d = 2|\Delta H|$) that also fixes the curvature to $-\frac{1}{4}$ on the Gaussian leaf. The load-bearing results (theorem 4.10: wall; theorems 4.18 and 4.19: hyperbolicity, $K = -1/(2cJ_0)$) hold for the *whole* family; changing the representative is a change of unit (theorem 4.23), which rescales the value, the distances and the curvature but preserves the sign, the ranking and the Gaussian extremum.

Change of cost unit. We treat the level e and the relief U as the same kind of quantity — cost densities —, so that a change of unit rescales them together: for $\lambda > 0$, $(e, U) \mapsto (\lambda e, \lambda U)$ (on the family $\mathcal{F} = \{cJ\}$, $c \mapsto \lambda c$ and $e \mapsto \lambda e$).

Theorem 4.23 (gauge invariance of cost). *Under the change of cost unit $(e, U) \mapsto (\lambda e, \lambda U)$, $\lambda > 0$, the cost geometry (3.1) transforms by homothety, for all $e \geq 0$ and all $U \in \mathcal{C}$:*

$$\tilde{g}_{e,U} \mapsto \lambda \tilde{g}_{e,U}, \quad d_e \mapsto \sqrt{\lambda} d_e, \quad K \mapsto K/\lambda$$

(the last on the location-scale leaves, of dimension 2). Thus gauge-invariant: the sign of K and hyperbolicity; the finiteness or infiniteness of the d_e -distances — the wall (theorem 4.10) — and well-posedness; the traces of the geodesics and the ratios of distances; and the eikonal proportionality $U = cJ$ (theorem 4.15). At $e = 0$, where K is constant (theorem 4.18), one further gains the ranking of the leaves by curvature and its Gaussian extremum (theorem 4.19), whose ratio $K(\text{gauss})/K(\varphi_0) = J_0$. Depending on the unit (scale images): every absolute distance, the price of a nat $\kappa = 1/\sqrt{2c}$, and the value $K = -1/(2cJ_0)$ (i.e. $-1/(4J_0)$ at the representative $c = 2$, and $-\frac{1}{4}$ for the Gaussian $J_0 = 1$).

Proof. Under $(e, U) \mapsto (\lambda e, \lambda U)$, $\tilde{g}_{e,U} = 2(e + U) g_{\mathcal{W}_2}$ becomes $2(\lambda e + \lambda U) g_{\mathcal{W}_2} = \lambda \tilde{g}_{e,U}$ (all $e \geq 0$). A length $\ell_e = \int \sqrt{2(e + U)} |\dot{\gamma}|$ is thus multiplied by $\sqrt{\lambda}$, whence $d_e = \inf \ell_e \mapsto \sqrt{\lambda} d_e$; the infimum being over the same set of curves, the minimizing traces, the ratios of distances and the finiteness/infiniteness of the lengths — hence the wall (theorem 4.10) — are unchanged. On a conformal leaf $\tilde{g} = f(d\mu^2 + d\sigma^2)$, $K = -\Delta(\ln f)/(2f)$, and $f \mapsto \lambda f$ gives $\Delta \ln(\lambda f) = \Delta \ln f$ with a denominator $2\lambda f$, whence $K \mapsto K/\lambda$ pointwise. A positive scalar factor preserves the sign and curvature ratios; at $e = 0$, where K is constant (theorem 4.18), it moreover preserves the *ranking* of the leaves, whence the Gaussian extremum (theorem 4.19). Finally, for an eikonal (hence continuous, $U = cJ$) relief — where the slope identity of theorem 4.15 applies —, the eikonal slope $\|\nabla_{\tilde{g}_{0,U}} H\| = \sqrt{J}/\sqrt{2U} = \kappa$ becomes $\kappa/\sqrt{\lambda}$, a nonzero constant: the proportionality $U = cJ$ (theorem 4.15) is preserved (its value κ , in turn, is gauge). $\square$

Box 1 (Algorithmic application). Three objects of machine learning fall into this framework — two reduce to it exactly, one shares its rule (Fisher = cost of change) —, separated by σ : inside ($\sigma > 0$) a belief is revised at finite cost; at the boundary ($\sigma = 0$, the wall ∂_∞) one must retrain.

1. Kalman — interior point. In the linear-Gaussian regime, the belief of a scalar Kalman filter (Kalman, 1960) is the *exact* posterior $b_t = \mathcal{N}(\mu_t, \sigma_t^2)$, a point of the Gaussian leaf (figure 2 (a)); its recurrence moves within it — prediction *opens* the belief, correction *tightens* it. This point is at finite cost distance from any other interior belief (corollary 5.2), hence revisable at finite cost by each datum.

2. The point predictor — the wall. A point estimate $\hat{x}$ (variance-free regression, arg max, collapsed output) is modeled as the Dirac $\delta_{\hat{x}}$, a point of ∂_∞ ($H = -\infty$): a point estimate is the MAP (maximum a posteriori) — the mode of a posterior whose spread has collapsed —, the quantity training-as-inference minimizes (MacKay, 2003, p. 493). By theorem 4.10(i), it is at *infinite* cost distance from any interior point: no continuous revision exists, one can only *retrain*. The overconfident softmax is its practical instance, *near* the wall ($H \rightarrow -\infty$ without reaching it); keeping the predictive *moderated* (uncertain) is the well-behaved regime (MacKay, 2003, p. 502).

3. EWC — the same rule, carried to weights. *Elastic Weight Consolidation* (Kirkpatrick et al., 2017) penalizes the displacement of weights by $\sum_i F_i (\theta_i - \theta_i^*)^2$, F_i the diagonal of the Fisher information: the cost of changing a parameter is proportional to its *certainty* — the same principle as the cost metric, carried from beliefs to parameters. *Scope*: this F_i is a Fisher–Rao tensor on the weights (principle 1), whereas the J of the framework is the slope of $-H$ on transport — a kinship of principle, not a geometric identity.

Remark 4.24 (invariants and choice of unit). The wall, the eikonal and Stam rigidity are the invariants of this change of unit, and no proven conclusion depends on an absolute cost. The value $-\frac{1}{4}$ rests on two conventions: the gauge $c = 2$ and the standardization $\text{Var}(\varphi_0) = 1$, which gives $J_0 = 1$; the invariant content is the sign, the ranking, the extremum and the ratio $K(\text{gauss})/K(\varphi_0) = J_0$. Thermodynamics fixes the unit of cost (remark 4.17) and motivates this framework, without entering into a proof. In entropic terms this is an *orthogonality*: the entropy is *what* the cost measures; the cost gauge c is the *unit* in which it is measured (the quadratic exponent $p = 2$ itself being *forced*, proposition 3.5, not a unit) — the value $-\frac{1}{4}$ being one image of that gauge.

Proposition 4.25 (admissibility of $\mathcal{F}$). $\forall c > 0 : cJ \in \mathcal{C}$: *positivity and τ -l.s.c. by assumption 9; global domination $cJ \geq \varepsilon J$ with $\varepsilon = c$. Hence $\mathcal{F} \subset \mathcal{C}$ and all the machinery (Section 4.2) and the wall (theorem 4.10) apply.*

5 Consequences: a well-posed inference

The wall (theorem 4.10) pushes the boundary of certainties ∂_∞ to infinite d_e -distance: every curve approaching it has a cost length ℓ_e that diverges. Three consequences follow: a minimizer exists without collapsing, the cost bounds the entropy variation, and reaching a precision has a geometric cost floor.

Corollary 5.1 (existence and attainability of the minimizer). *By theorem 4.10, the cost length ℓ_e diverges on approach to ∂_∞ : it is coercive on the candidate class, so no minimizing sequence of finite cost approaches the boundary. Existence then follows by the direct method (theorem 4.5), with no appeal to geodesic completeness: the topology used is the narrow (τ) one, in which the sublevels are compact (assumption 6), not the d_e -metric topology (in which $(\mathcal{P}_2, \mathcal{W}_2)$, refined by*

$d_e \geq \sqrt{2e} \mathcal{W}_2$ — since $\tilde{g}_{e,U} = 2(e+U) g_{\mathcal{W}_2} \geq 2e g_{\mathcal{W}_2}$ gives $\ell_e \geq \sqrt{2e} \text{length}_{\mathcal{W}_2} \geq \sqrt{2e} \mathcal{W}_2$ —, is not locally compact). We deliberately avoid the metric Hopf–Rinow route (assumption 8, a contrasted standard fact, not a tool here) precisely because the cost-metric space is not locally compact, so its hypotheses fail; the direct method on the narrow topology is what carries the argument. A minimizing sequence then stays in a sublevel $\{m_2 \leq M\}$, τ -sequentially compact; Arzelà–Ascoli extracts a limit, which the lower semicontinuity of the action (theorem 4.5; Buttazzo et al., 1998) selects as a minimizer in the interior of the domain (theorem 4.5). The minimizer exists and does not collapse onto certainty.

Corollary 5.2 (Lipschitz control of the cost in entropy). *By theorem 4.10, the boundary is at infinite d_e -distance: no curve of finite cost length reaches it, and every pair of beliefs in the interior of $\mathcal{R}$ (where J is locally finite, so ℓ_e is finite along a joining segment) stays at finite d_e -distance. The Lipschitz property of the entropy ($|\Delta H| \leq \frac{1}{\sqrt{2\varepsilon}} d_e$, theorem 4.10(i)) bounds the entropy variation along any curve by its cost length: each unit of length can lower the entropy by at most $\frac{1}{\sqrt{2\varepsilon}}$. This control bears only on the entropic term: the data-attachment and interaction contributions are not bounded by the wall.*

Corollary 5.3 (geometric cost floor). *By theorem 4.10(i), every curve joining two beliefs whose entropies differ by ΔH has a d_e -length at least $\sqrt{2\varepsilon} |\Delta H|$. Reaching a given precision (target variance) thus has a geometric cost floor. The bound is sharp: it is attained in the eikonal limit $U = \varepsilon J$ at $e = 0$, where $d_0 = \sqrt{2\varepsilon} |\Delta H|$ (remark 4.17) along the $\mathcal{W}_2$ -gradient flow of $-H$ that saturates assumption 5; for $e > 0$ or $U > \varepsilon J$ it is strict. On the Fisher family $\mathcal{F} = \{cJ\}$, where $\varepsilon = c$ (proposition 4.25), $d_e \geq \sqrt{2c} |\Delta H|$ diverges with $|\Delta H|$ at the wall. (The energetic reading of this floor — its proportionality to the Landauer energy at the boundary — is deferred to remark 4.17, tier-P, outside the proof.) This floor is purely geometric, in nats; its value depends on the cost unit, and its divergence at certainty is invariant of it (theorem 4.23).*

Existence without uniqueness. Completeness and coercivity give the existence of a minimizer, not its uniqueness. The latter would require the geodesic convexity of the objective in the cost metric, not established in general (Section 6). The framework establishes well-posedness *for existence and stability*; uniqueness remains open — the non-convexity of possible interaction terms, and the multiplicity of minima it may induce, are left open.

6 Discussion: scope and limits

Three results are established: a wall, an eikonal equivalence, a curvature bounded below by Stam. It remains to situate them — with respect to what stays conjectural and to what we do not claim (below); the positioning relative to neighboring geometries is treated together with the related work (Section 7).

What remains conjectural

Three statements bound the scope of the theorems; we fix their exact status, without broadening it.

Remark 6.1 (C1 — beyond the conformal class). theorem 4.10(i) (sufficiency) holds for every $U \in \mathcal{C}$; theorem 4.10(ii) (necessity) is proved on the power family $\{cJ^\alpha\}$. **Conjecture C1.** (a) Necessity extends to every $U \in \mathcal{U}$ (any U not dominating J near ∂_∞ leaves the boundary at finite distance). (b) Beyond conformal deformations (definition 3.6), the characterization remains open. *We never claim uniqueness outside the conformal class. (The choice of exponent $p = 2$ within the transport family is settled separately by proposition 3.5 and is orthogonal to this conjecture.)*

Remark 6.2 (C2 — curvature beyond the Gaussian: resolved on location-scale). **Resolved (location-scale).** Beyond the Gaussian leaf, theorems 4.19 and 4.23 establish that on every 1D

location-scale family the curvature, at $e = 0$, equals $-1/(2cJ_0) < 0$: *negativity is robust*, and the extremal value is attained by the Gaussian (Stam, $J_0 = 1$). **Conjecture C2 (what remains open)**. (a) Negativity near ∂_∞ persists on 2D families that are *not* location-scale (curved $\mathcal{W}_2$ base). (b) In dimension ≥ 2 / on all of $\mathcal{P}_2$, the *ambient sectional* curvature of $\tilde{g}_{e,2J}$ is negative near ∂_∞ — a heavier object: Otto computation + *dimension-dependent* conformal transformation. *Tooling note*: the curvature-dimension condition $CD(K, N)$ (here K is a generic Ricci lower bound, not the Gauss curvature) (Lott and Villani, 2009; Sturm, 2006) bounds the Ricci of the *base* $(\mathcal{P}_2, \mathcal{W}_2)$ via convexity of the entropy; this is *not* the curvature of the cost metric $\tilde{g}$ (a distinct conformal object) — $CD(K, N)$ therefore applies only to part (b), not to theorem 4.19. *We do not conflate the intrinsic family curvature (theorem 4.19) and the ambient sectional one.*

Remark 6.3 (C3 — global existence on $\mathcal{P}_2$). theorem 4.5 gives existence for the action; the existence of the minimizer of an estimate on all of $\mathcal{P}_2(\mathbb{R}^n)$ (mass escape) requires concentration-compactness (Lions, 1984), beyond the reach of a properly proved statement as it stands. *Status*: scoped; the interior minimizer guaranteed by the wall (theorem 4.10) suffices for fixed endpoints / coercive fidelity.

Acknowledged limits

We claim no more than what is established: the curvature *reads* the Fisher information, it does not ground it; the value $-\frac{1}{4}$ is the image of a unit (remark 4.22); “well-posed” covers only existence and stability; and we characterize only a *family*, within the *conformal* class. Three of these — general necessity, universality of the curvature, global existence on $\mathcal{P}_2$ — are the conjectures scoped above (remarks 6.1 to 6.3); uniqueness of the minimizer is the open point of the “Existence without uniqueness” paragraph (Section 6).

7 Related work

Each building block is cited where it is used; we gather here the boundary between what is acquired and what is contributed, complementing the three positioning remarks (remarks 7.1 to 7.3). The contribution amounts to two moves: the eikonal characterization $U = cJ$ (theorem 4.15) and the assembly into a cost metric with a wall; the rest is borrowed.

Geometry of inference. Information geometry (Chentsov, 1972; Amari, 2016) equips beliefs with the Fisher–Rao metric, the only one invariant under statistical morphisms. Otto’s calculus (Otto, 2001) makes the complementary move: it endows $\mathcal{P}_2$ with a Riemannian structure of distance $\mathcal{W}_2$, where the heat flow is the gradient flow of entropy (Jordan et al., 1998; Ambrosio et al., 2008) and where $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ (Villani, 2009) — the identity that makes J the *slope* of $-H$ (assumption 4), not a tensor — the metric form of de Bruijn’s classical identity $\frac{d}{dt}H(X_t) = \frac{1}{2}J(X_t)$ along the heat flow (Cover and Thomas, 2006, Th. 17.7.2). The HWI (entropy–Wasserstein–Fisher) inequality (Otto and Villani, 2000) already relates our objects in one line. Our framework characterizes *another* class, the conformal deformations of transport, whose curvature signature $-\frac{1}{4}$ separates it from Fisher–Rao (remark 7.1).

Thermodynamics of information. Erasing one nat dissipates at least $k_B T$ (Landauer, 1961; Bennett, 1982); the thermodynamics of information turned this into an exact bookkeeping (Parrondo et al., 2015) — extractable work bounded by the information acquired (Sagawa and Ueda, 2010), and an *energetic* reading of geodesics where the path of minimal dissipation is optimal transport (Aurell et al., 2011; Dechant and Sakurai, 2019). This is the currency, nat for nat, and the legitimacy of treating an inference device as a thermodynamic system; the energetic optimality of geodesics is their theorem, not ours. Our cost is *derived* from the right price.

Finite-time dissipation. Benamou–Brenier (Benamou and Brenier, 2000) gives $\mathcal{W}_2^2$ a kinetic reading; stochastic thermodynamics turns it into a price — the dissipation of a diffusion driven from p_0 to p_1 in time T is bounded below by $\mathcal{W}_2^2(p_0, p_1)/T$ (Dechant and Sakurai, 2019), and the thermodynamic length (Sivak and Crooks, 2012) is its linear regime; a second family bounds the *rates* by the Fisher of the path (Ito, 2018). These works *measure* the dissipation on a given geometry; they do not re-metrize the space of beliefs nor place a wall in it, and their Fisher is that of the path, not of the state $J(p)$ (remark 7.2).

Fisher cost in learning. This is a direct link. Elastic weight consolidation (Kirkpatrick et al., 2017) fights forgetting by penalizing the displacement of a belief by a *squared Fisher distance* toward a reference state, derived from a Laplace approximation of the posterior (Huszár, 2018) — Laplace’s method, the Gaussian posterior near the mode (MacKay, 2003, p. 341, and in learning p. 501) —: on the Gaussian leaf, the same object as our maintenance cost, but *static* — a frozen anchor — where the maintenance cost is dynamic. Still et al. (2012) price *non-predictive* information under driving and Kolchinsky and Wolpert (2018) ground semantic information thermodynamically; here even useful information has a permanent maintenance cost, because in our setting the world diffuses. The natural gradient (Amari, 2016) preconditions by the Fisher, a relative of well-posedness (corollary 5.1).

Mixed Wasserstein–Fisher geometries, and positioning. von Renesse (2012) reads the Schrödinger equation as a Newton law on $\mathcal{P}_2$ with potential the Fisher, and Conforti and Pavon (2018) classify transport, Schrödinger bridge and Madelung fluid as extremals of actions that differ only by the *sign* of a Fisher term; to be distinguished from the Wasserstein–Fisher–Rao metric (Chizat et al., 2018), which interpolates *additively* by varying the mass, the opposite of our conformal deformation at mass 1. The conformal (multiplicative) reweighting of the transport metric itself has a precedent: Ambrosio and Santambrogio (2007) multiply the Wasserstein metric by a scalar factor (a power of an L^q norm of the density) to favour spreading, with no curvature; here the factor is the Fisher information, from which the wall, the curvature and the Stam extremality follow. Concurrent works (2024–2026) treat neighboring facets without covering this assembly: Okanohara (2026) bounds the dissipation of learning by an ensemble Wasserstein distance (transport of models, a relative of that of Dechant and Sakurai (2019), where the maintenance cost bears on the *state* of Fisher $J(p)$), Melo et al. (2025) study the entropy production tied to Fisher information in stochastic thermodynamics, of which Ito and Sagawa (2016) gives the Bayesian-network setting, and Hyland and Albarracin (2025) price the cost of *changing* belief, where the wall (theorem 4.10) prices that of *holding* it. To our knowledge, the eikonal characterization $U = cJ$ and the metric inaccessibility of certainty as a wall on the conformal class have not been formulated as such.

Positioning

The characterization is read first by contrast, opposing three currencies — Fisher–Rao, bare transport and honest cost — on the same location-scale leaf. The principal contrast is with Fisher–Rao, and it sharpens into a thesis *dual* to Čencov’s:

Fisher–Rao is the canonical geometry of distinguishability (Čencov’s uniqueness); the conformal, mass-preserving, Fisher-reweighted transport metric is, we conjecture in general, the canonical geometry of revision cost (theorem-grade on the power family; general necessity is Conjecture C1, remark 6.1).

The two uniqueness statements answer different questions — how distinguishable two beliefs are, versus what it costs to move between them —; as with Čencov, the uniqueness here is relative to its class (remark 7.1). table 2 makes the contrast checkable at a glance: the four *discriminating*

Table 2: What each arena can and cannot state. Rows: capabilities the characterization needs; columns: candidate arenas (WFR: Wasserstein–Fisher–Rao). ✓ available; × absent; ~ present but in a form that changes the object. The four *discriminating* rows (bold) are × everywhere but in $\mathcal{W}_2$. The wall’s discrimination is theorem-grade on the power family $\{cJ^\alpha\}$; general necessity is Conjecture C1 (remark 6.1).

Required capability	Fisher–Rao	KL	$W_{p \neq 2}$	WFR	$\mathcal{W}_2$ (ours)
Genuine metric (distance, geodesics)	✓	×	✓	✓	✓
Riemannian structure + curvature	✓	×	×	✓	✓
J as slope of $-H$ (Otto)	×	×	×	×	✓
Wall discriminates on U	×	×	×	×	✓
Eikonal $U=cJ$ has content	×	×	×	×	✓
Stam $\Rightarrow$ Gaussian extremum	×	×	×	×	✓
Mass conserved (probability = 1)	✓	✓	✓	×	✓
Dissipation grounding (physical cost)	×	×	~	~	✓

Why the crosses (Fisher–Rao unless noted): KL is asymmetric (no metric); $W_{p \neq 2}$ has no Riemannian/Otto structure; Fisher–Rao makes J the tensor, not the slope of $-H$, so the eikonal has nothing to select (no Otto identity); the wall is free (the Dirac is already at infinite distance, nothing to discriminate); and Stam bounds the position information, whereas Fisher–Rao location-scale curvature reads the scale information (symmetric base; see remark 4.21), which Stam does not bound. WFR varies the mass, changing the object.

capabilities live in $\mathcal{W}_2$ alone. The three remarks that follow unfold the argument, each against an established program.³

Remark 7.1 (vs information geometry (Amari–Čencov)). Čencov characterizes Fisher–Rao as the unique metric invariant under statistical morphisms (Čentsov, 1972; Amari, 2016). Our characterization is of analogous nature but in another class: conformal deformations of transport (Wasserstein), where J is the slope of $-H$ (assumption 4) and not the metric tensor — the demotion (principle 1) read once more, here in the positioning itself. Distinctive signature: curvature $-\frac{1}{4}$ (representative) vs $-\frac{1}{2}$ of Fisher–Rao (Atkinson and Mitchell, 1981; Costa et al., 2015). On location-scale families, Fisher–Rao is governed by the scale information (Gaussian $-\frac{1}{2}$, with no link to Stam) (Atkinson and Mitchell, 1981; Costa et al., 2015) and bare transport has curvature ≥ 0 (Takatsu, 2011), where the honest cost makes the position information J_0 carry the curvature, J_0 being bounded below by Stam (theorem 4.19 and remark 4.21) — the negativity arises from the conformal deformation, not from bare transport. Deforming a base metric to negative curvature on location-scale families is a known template — warped Fisher–Rao (Said et al., 2017), the λ -deformation to a conformal Hessian metric of constant curvature (Zhang and Wong, 2022), the Hessian metric of an information functional on $\mathcal{P}_2$ (Li, 2021). The closest of these, Li (2021), builds the Hessian of an information functional (a full second-derivative tensor) on $\mathcal{P}_2$, generically not conformal to $g_{\mathcal{W}_2}$; here the Fisher enters as a scalar conformal factor $2(e + U)g_{\mathcal{W}_2}$ on the transport metric, and none of Said et al. (2017); Zhang and Wong (2022); Li (2021) yields the inaccessibility wall, the eikonal selection $U = cJ$, or the Stam extremality. The same line separates us from the Wasserstein natural gradient and Wasserstein information matrix (Li and Montúfar, 2018; Li and Zhao, 2023), which pull the transport metric back as a tensor preconditioner on a finite-dimensional parameter space (with a Wasserstein–Cramér–Rao bound on location-scale families), and from the Wasserstein statistics of Amari and Matsuda (2024) on those same affine families, which use bare transport — no conformal J -reweighting — and so yield neither wall, nor negative curvature, nor Stam extremum. As with Čencov, the uniqueness (here of family) is relative to the class; we make it explicit.

Remark 7.2 (vs geometric thermodynamics (Ito)). Stochastic thermodynamics establishes the Fisher $\leftrightarrow$ Wasserstein bridge (Sivak and Crooks, 2012; Aurell et al., 2011; Ito, 2023): entropy

³The location-scale leaves are the noise models of M-estimators ($\varphi_0 \leftrightarrow \text{loss } \rho = -\log \varphi_0$); the curvature $K = -1/(2cJ_0)$ (theorems 4.19 and 4.23) orders them, the Gaussian (least squares, $J_0=1$) being the Stam extreme — a bridge to robust estimation (Huber, 1981) left for elsewhere.

production is bounded by $\mathcal{W}_2$ (a speed limit), J the excess entropy rate. These works *measure* dissipation on a given geometry; we *characterize* a class of conformal *cost* metrics with a wall and a curvature.

Remark 7.3 (vs Bayesian mechanics (Friston)). The free-energy principle carries “dissipative agent $\rightarrow$ geometry of belief” on a Fisher manifold (Friston, 2019; Sakthivadivel, 2022), without a wall, negative curvature driven by *position* information, or an eikonal $U = cJ$ on Wasserstein. This work’s contribution is that *assembly* — eikonal and wall on transport — more than each ingredient alone.

8 Perspectives

The preceding characterizations concern a *background* cost geometry: fixed by honesty (theorem 4.15) and determined up to a unit (theorem 4.23), the metric $\tilde{g}_{e,U} = 2(e + U) g_{\mathcal{W}_2}$ is a scene given in advance, where a belief moves like a test particle. A *self-consistent* geometry, sourced by the configuration of beliefs it carries rather than prescribed, could extend it. The displacement would then bear on the scene itself, beyond the cost unit fixed by gauge invariance (theorem 4.23); we borrow from general relativity, as an analogy, its lesson of background independence (Einstein, 1916). One ingredient already lends itself to this: in the kinetic action $\frac{1}{2}|\dot{\gamma}|_{\tilde{g}}^2 = (e + cJ)|\dot{\gamma}|_{\mathcal{W}_2}^2$, the factor $(e + cJ)$ would play the role of an *inertial mass* — the resistance to changing one’s mind, large near the wall (theorem 4.10). It remains to define the *active* mass of a belief — by which it could, in return, contribute to the geometry seen by other beliefs — and its coupling. The three statements below, in the conditional, mark out the way⁴; following the convention of the paper, they *motivate* and enter into no proof.

Remark 8.1 (C4 — active mass and inertial mass). In a self-consistent completion of the cost geometry — where the relief U would be sourced by the belief configuration —, the active mass of a belief would coincide with its inertial mass $\propto (e + cJ)$, already present as the factor of $|\dot{\gamma}|^2$ in $\tilde{g}$.

Remark 8.2 (C5 — toward an epistemic hysteresis). A nonconvex attractive interaction term between correlated beliefs would induce a multiplicity of minima; under a quasi-static dynamics, the belief would exhibit a *hysteresis*, whose loop width would grow with the inertial mass ($\propto e + cJ$) and the coupling. The two-route first-order transition signaled in theorem 4.7 would be its minimal instance.

Remark 8.3 (C6 — relativized honesty: toward a self-consistent geometry). Postulate 1 fixes a *globally* constant price per nat ($\kappa \text{ const} \Rightarrow U = cJ$). Relaxing it to a *local* price, $\kappa = \kappa(p) \Rightarrow U = c(p)J$, turns the cost unit into a *field* over the space of beliefs, keeping the pointwise proportionality $U \propto J$ (only the coefficient c becomes a field). This is the field-over-configuration counterpart of the self-consistent geometry of remark 8.1: relativizing loses uniqueness — $c(p)$ is a free function — unless a field equation determines it from the belief configuration, which is precisely the coupling left undefined in remark 8.1; hence a well-posed open problem, not a result. A natural source already present is the plausibility $\Pi \propto \nu \exp(-U)$ (definition 3.4). The heterogeneity it produces is *configurational* (cheaper and dearer regions of belief space, hence agent-dependent revision), not a dynamics in time: it leaves the static reading (remark 4.20) untouched.

The development of these directions — self-consistent geometry, interaction dynamics, hysteresis — belongs to separate work.

9 Conclusion

A finite agent’s belief is denied certainty twice (Section 3). On the conformal class, the wall — certainty pushed off to infinite distance — is *characterized* by the dominance of Fisher infor-

⁴C4, C5 and C6 follow the numbering style of conjectures C1–C3 (remarks 6.1 to 6.3); unlike those, which bound the scope of the established theorems, C4–C6 open directions outside the proof structure.

mation (sufficient unconditionally; necessary at the boundary, general case conjectured; theorem 4.10); there the eikonal selects the Fisher family $U = cJ$ (theorem 4.15), and the curvature $K = -1/(2cJ_0)$, over any location-scale family, is extremal at the Gaussian, by Stam rigidity (theorems 4.18 and 4.19). These three characterizations are the invariants of one and the same change of cost unit (theorem 4.23): thermodynamics fixes the unit, and the proven conclusions hold up to that unit. We thus obtain, within the conformal class, a *family* over which inference is well-posed, and reaching a precision has a floor that diverges at certainty. Its extension — a geometry sourced by the beliefs it carries — is sketched in Section 8.

A Proofs of the structural results

We prove here the class-machinery statements announced in Section 4.2 — the AM–GM inequality (lemma 4.3), the Maupertuis–Jacobi correspondence (theorem 4.4), the existence of geodesics (theorem 4.5), the energy law (proposition 4.6) and the equivalence optimal curve $\Leftrightarrow$ geodesic (theorem 4.7) —, valid for every candidate $U \in \mathcal{U}$; the regime clauses that bound them (duality gap $e > 0$, du Bois-Reymond lemma, compactness of the sublevels by Arzelà–Ascoli) enter explicitly in each proof.

Proof of lemma 4.3. $\frac{1}{2}a^2 + b - \sqrt{2b}a = \frac{1}{2}(a - \sqrt{2b})^2 \geq 0$, zero iff $a = \sqrt{2b}$. $\square$

Proof of theorem 4.4. (i) lemma 4.3 with $a = |\dot{\gamma}|$, $b = e + U$, integrated. (ii) reparametrize at constant $\mathcal{W}_2$ speed L (Ambrosio et al., 2008, lem. 1.1.4), set $t(s) = \int_0^s L/\sqrt{2(e + U(\hat{\gamma}_r))}dr$ (integrand $\leq L/\sqrt{2e}$, > 0 a.e.); the equality case of lemma 4.3 concludes. (iii) equality in (i) = pointwise equality case. $\square$

Proof of theorem 4.5. Minimizing sequence $\mathcal{A}_T(\gamma^k) \leq C$; since $U \geq 0$, lemma 4.1(a) gives $\mathcal{E}(\gamma^k) \leq 2\mathcal{A}_T(\gamma^k) \leq 2C$, whence the Hölder bound $\mathcal{W}_2(\gamma_s^k, \gamma_t^k) \leq \sqrt{\mathcal{E}(\gamma^k)}|t - s| \leq \sqrt{2C}|t - s|$. A uniform second-moment bound follows: as $\sqrt{m_2(\mu)} = \mathcal{W}_2(\mu, \delta_0)$, $\sqrt{m_2(\gamma_t^k)} \leq \mathcal{W}_2(\gamma_t^k, p_0) + \mathcal{W}_2(p_0, \delta_0) \leq \sqrt{2CT} + \sqrt{m_2(p_0)} =: \sqrt{M}$, so the curves stay in the τ -compact $\{m_2 \leq M\}$ (assumption 6), on which τ is metrized by the Dudley metric $\beta_0 \leq \mathcal{W}_1 \leq \mathcal{W}_2$ ($\mathcal{W}_1$ the 1-Wasserstein distance); the $\mathcal{W}_2$ -Hölder bound is a fortiori β_0 -equicontinuity, so Arzelà–Ascoli extracts a τ -limit γ . By τ -l.s.c. of $\mathcal{E}$ (lemma 4.1(c)) and Fatou for $\int U$ (U l.s.c.), one gets $\mathcal{E}(\gamma) \leq 2C < \infty$ (hence $\gamma \in AC^2$, lemma 4.1(b)) and $\mathcal{A}_T(\gamma) \leq \liminf_k \mathcal{A}_T(\gamma^k) = \Phi(T)$. $\square$

Proof of proposition 4.6. Inner variations: for $u : [0, T] \rightarrow (0, \infty)$, $u, 1/u \in L^\infty$, $\int u = T$, with $\theta_u(t) := \int_0^t u$, $\gamma_t^u := \gamma_{\theta_u(t)}^*$ is admissible and $\mathcal{A}_T(\gamma^u) = \int [\frac{1}{2}|\dot{\gamma}^*|^2/u + Uu] =: F(u)$. F convex, $u \equiv 1$ optimal. $V(s) := \inf\{F : \int u = s\}$ convex; $\text{dom } V = (0, \infty)$ (for every $s > 0$, $u_s := s/T$ gives $\int u_s = s$ and $F(u_s) = \frac{T}{s} \int \frac{1}{2}|\dot{\gamma}^*|^2 + \frac{s}{T} \int U(\gamma^*) < \infty$), so T is interior there and $\partial V(T) \neq \emptyset$; taking $-e \in \partial V(T)$, the first variation $\int [-\frac{1}{2}|\dot{\gamma}^*|^2 + (U + e)]h = 0$ ($\forall h \in L^\infty$) gives (du Bois-Reymond) the stationarity, and the convexity of $u \mapsto \frac{1}{2}|\dot{\gamma}^*|^2/u + (U + e)u$ ($\partial_u^2 \geq 0$) makes it the pointwise minimizer. $\square$

Proof of theorem 4.7. The energy law gives the pointwise equality of lemma 4.3, hence $\mathcal{A}_T(\gamma^*) + eT = \ell_e(\gamma^*)$ and the optimality of the trace for ℓ_e at parameter $T_e = T$. Write $G(x) := \ell_x(\gamma^*) - xT$ and $D(x) := \Psi(x) - xT$; since $\Psi \leq \ell(\gamma^*)$ pointwise, $D \leq G$. G is strictly concave ($G'' < 0$) and $G'(e) = \int |\dot{\gamma}^*|/\sqrt{2(e + U)} - T = 0$ by the energy law ($e > 0 \Rightarrow |\dot{\gamma}^*| = \sqrt{2(e + U)} > 0$ a.e.), so e is the unique maximizer of G , with $G(e) = \ell_e(\gamma^*) - eT$. D is concave with $D(x) \rightarrow -\infty$ as $x \rightarrow +\infty$, so $\sup_x D$ is attained, say at x^* ; exact duality gives $\Phi(T) = D(x^*) \leq G(x^*) \leq G(e)$, whence, with $\Phi(T) = \mathcal{A}_T(\gamma^*) = \ell_e(\gamma^*) - eT = G(e)$, equality throughout and, by strict concavity, $x^* = e$. Therefore $\Psi(e) = D(e) + eT = \ell_e(\gamma^*)$ (global geodesic). Converse: theorem 4.4(ii). $\square$

Table 3: Dependency table (authoritative). Provenances: **(P)** physical/interpretive (motivates, never a proof); **(E)** borrowed (standard theorem cited); **(D)** proved here, in pure metric geometry.

Statement	Tier	Direct logical dependencies
P: physical (Landauer/Bekenstein)	(P)	— (<i>motivates, dashed: Landauer fixes the unit; the dissipation floor $\mathcal{W}_2^2/T$ corroborates Postulate 0; operational constraints, remark 3.3</i>)
P0, P1 (postulates)	posed	— (<i>defended choices; P1 presupposes P0, via Char. II; enter as hypotheses, no incoming edge</i>)
A1, A2; B1–B7	setting/ (E)	—
A3 arena (assumption 3)	setting	Postulate 0 (transport type); proposition 3.5 ($p=2$)
lemma 4.3, lemma 4.1	(D)	—; B3
lemma 4.2	(E)	B1, B2
Maupertuis (theorem 4.4)	(D)	lemma 4.3
existence (theorem 4.5)	(D)	lemma 4.1, B3
energy law (proposition 4.6)	(D)	theorem 4.5
geodesic (theorem 4.7)	(D)	theorem 4.4, proposition 4.6
Char. I wall (theorem 4.10)	(D)	definition 4.8, lemma 4.2, B4, theorem 4.5 (<i>B1 only transitively, via lemma 4.2; carI uses only the B2 inequality</i>)
Char. II eikonal (theorem 4.15)	(D)	B1 (Otto/A2), definition 3.6 (conformity), definition 4.13 (eikonal condition, used as hypothesis — the content P1 posits; Char. II is a conditional equivalence, not assuming P1 true, cf. proposition B.1)
admissibility $\mathcal{F}$ (proposition 4.25)	(D)	B6
hyperbolicity (theorem 4.18)	(D)	definition 4.14, B4
Stam extremum (theorem 4.19)	(D)	B7, B6 (l.s.c. dual); Stam extremality (Stam, 1959; CT Ch. 17)
$p = 2$ forced (proposition 3.5)	(D)	B7, B4, Jordan–von Neumann (cited)
representative $2J$ (remark 4.22)	(D)	theorem 4.15, theorem 4.18, theorem 4.19
setting defs.	setting	definitions 3.6, 4.8, 4.9, 4.13 and 4.14 — (<i>e and the “2” enter as a unit; cf. theorem 4.23</i>)
gauge invariance (theorem 4.23)	(D)	definition 3.6, theorem 4.19
finiteness cor. (corollary 4.11)	(D)/(P)	theorem 4.10; <i>motivates: P</i> (dashed)
honesty of the price (remark 4.17)	(D)/(P)	theorem 4.10, theorem 4.15; <i>motivates: P</i> (dashed)
base (definition 3.1, definition 3.2)	setting	— (<i>object: H, J, ∂_∞</i>)
existence (corollary 5.1)	(D)	theorem 4.10, theorem 4.5, assumption 6
cost–entropy (corollary 5.2)	(D)	theorem 4.10(i)
cost floor (corollary 5.3)	(D)/(P)	theorem 4.10(i); <i>motivates: P</i> (dashed)

B Dependency table and logical independence of the proofs

table 3 records, statement by statement, the authoritative direct logical dependencies; it is the only place where the provenances P (physical/interpretive), E (borrowed) and D (proved) are labeled, following the convention recalled in the table’s note. The acyclicity proposition below establishes that no proved statement depends on a physical statement.

Proposition B.1 (acyclicity of dependencies: no proved statement depends on a physical statement **(D)**). *Let $\rightarrow$ denote direct logical dependence (col. 3 of table 3, which is authoritative; set-theoretic reading: P outside the universe). Here **(P)** physical (Landauer/Bekenstein, dissipation floor) is not the postulates P0, P1: the latter are posed hypotheses (setting tier) and, like every setting item, may have outgoing edges — promoting the eikonal to Postulate 1 adds the edge $P1 \rightarrow \text{Char. II}$ (which later **(D)** items, e.g. remarks 4.17 and 4.22, do consume); this is harmless*

because $P1$ is a posed setting-tier hypothesis, not the **(P)** physical statement P . The firewall below concerns **(P)** alone: part (ii) forbids only solid edges carrying P , and no col. 3 entry lists P (it appears only in the dashed “motivates” clause). Then: (i) the graph on $\{P\} \cup \{\text{setting}\} \cup \{\text{borrowed results (E)}\} \cup \{\text{statements (D)}\}$ is acyclic (stratification by rank: each solid edge goes from a lower rank to a higher rank); (ii) $\forall D \text{ (D)} : P \notin \text{cl}_{\rightarrow}(D)$: P has no outgoing solid edge (the “motivates” col. = dashed edges only), so every D is proved from setting + borrowed results alone. Each edge of table 3 has been traced back in the body of the corresponding proof (finite verification).

Proof. By inspection of table 3: no “logical dependencies” column contains P ; the rank ($\text{rank}(D) = 1 + \max\{\text{rank}(x) : x \rightarrow D\}$) is well-defined and finite, whence acyclicity; induction on the rank for $P \notin \text{cl}_{\rightarrow}(D)$. $\square$

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